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Continuous Optimization

A causation-based framework for pricing and cost allocation of energy, reserves, and transmission in modern power systems

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ABSTRACT

The increasing vulnerability of power systems has heightened the need for operating reserves to manage contingencies such as generator outages, line failures, and sudden load variations. Unlike energy costs, driven by consumer demand, operating reserve costs arise from addressing the most critical contingencies—prompting the question: how should these costs be allocated through efficient pricing mechanisms? As an alternative to previously reported schemes, this paper presents a new causation-based pricing framework for electricity markets based on contingency-constrained energy and reserve scheduling models. Major salient features include a novel security charge mechanism along with the explicit definition of prices for up-spinning reserves, down-spinning reserves, and transmission services. These features ensure more comprehensive and efficient cost-reflective market operations. Moreover, the proposed nodal pricing scheme yields cost recovery and system revenue adequacy based on the cost-causation principle. Numerical results from three case studies illustrate the performance of the proposed pricing scheme.

1. Introduction

The optimization of power system operation and planning has been a central theme of research for more than half a century (Kaya et al., 2026). Over this time, a core and enduring topic in this field stands out, namely ensuring system security—the ability to withstand contingencies such as generator outages, line failures, or sudden load variations without involuntary load shedding (Arroyo & Galiana, 2005). Although security has been a consistent priority for the power industry, the challenge has increased significantly in recent years. In modern power systems, besides energy, ancillary services have become indispensable for meeting diverse security requirements. Among these, operating reserves (ORs) are crucial for maintaining system security by balancing supply and demand. The transition towards deregulated markets and the large-scale integration of renewable generation have introduced new dynamics and uncertainties, making the traditional security problem more complex and critical than ever (Kaya et al., 2026). It is worth emphasizing that the increasing penetration of non-synchronous, inverter-based generation (e.g., wind and solar) has led to a significant reduction in overall system inertia. This physical inertia, traditionally provided by the rotating mass of conventional generators, is critical for grid stability as it naturally dampens frequency fluctuations. In low-inertia systems, a sudden contingency causes a much faster and deeper frequency deviation, thereby requiring larger and faster-acting ORs to maintain system security. Compounded by a growing exposure to natural hazards, these new dynamics have heightened existing vulnerabilities and amplified the role of ancillary services. Consequently, the costs of ORs represent an increasing share of operational costs (Badesa et al., 2025; Matamala et al., 2024; Shi et al., 2023).

Despite the increasing reliance on ancillary services, existing electricity markets have failed to establish adequate frameworks to meet the evolving challenges of modern power systems (Billimoria et al., 2020). Markets generally follow either joint or sequential designs for energy and reserve clearing (Galiana et al., 2005; Ribeiro et al., 2023). Joint markets, like those run across the United States (Federal Energy Regulatory Commission (FERC), 2011), co-optimize energy and reserves simultaneously. Conversely, sequential markets, common in Europe, clear energy and reserves separately. While sequential markets aim to optimize reserves as market products, they often rely on *ad hoc* allocation of transmission capacity for reserves, leading to inefficiencies and suboptimal outcomes (European Union Agency for the Cooperation

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of Energy Regulators (ACER), 2026). In contrast, joint markets typically achieve higher social welfare by better capturing the interdependencies between these products, as evidenced by several studies (Aganagic et al., 1998; Ahunbay et al., 2024; Arroyo & Galiana, 2005; Badesa et al., 2023; Bouffard et al., 2005; Byers & Hug, 2023; Chen et al., 2014; Galiana et al., 2005; Gan & Litvinov, 2003; Karangelos & Bouffard, 2012; Kirschen & Strbac, 2019; Morales et al., 2012; Wang & Chen, 2021; Wang et al., 2009; Wong & Fuller, 2007; Wu et al., 2004). Unfortunately, notwithstanding the significant improvements brought about by Independent System Operators (ISOs) in terms of coordination, transparency, and the centralized co-optimization of energy and ancillary services, many joint markets rely on static, exogenously defined reserve requirements, such as those in PJM (PJM Interconnection, L.L.C., 2025) and CAISO (California ISO, 2019). The primary motivation for these static methods lies in their simplicity. However, such simpler models fail to reflect the endogenous relationship between reserve needs and the energy dispatch under different system conditions. Moreover, reserve deliverability during contingencies is often overlooked, leading to *ad hoc* solutions like zonal divisions with separate reserve requirements and prices (Shi et al., 2023).

Another persistent challenge lies in the cost allocation of reserves. Most ISOs allocate reserve costs proportionally among load-serving entities based on energy consumption, which raises concerns about fairness and efficiency (Badesa et al., 2025; Matamala et al., 2024; Shi et al., 2023). Proportional allocation often results in cross-subsidies, as it fails to link costs directly to the entities responsible for reserve requirements. As far as the authors are aware, among existing markets, the cost-causation principle for reserve cost allocation is solely applied by the Australian Energy Market Operator (AEMO) across its two independent systems: the Wholesale Electricity Market (WEM) and the National Electricity Market (NEM). However, while the WEM employs a sequential approach that better aligns costs with responsible entities, the NEM still relies on proportional allocation (Australian Energy Market Operator (AEMO), 2023), perpetuating inefficiencies.

In light of these challenges, numerous studies have proposed deterministic frameworks, such as those relying on the $N-1$ security standard, as well as probabilistic approaches to improve reserve pricing and scheduling (Ahunbay et al., 2024; Arroyo & Galiana, 2005; Badesa et al., 2023; Bouffard et al., 2005; Byers & Hug, 2023; Galiana et al., 2005; Karangelos & Bouffard, 2012; Martin & Fanzeres, 2023; Mays et al., 2021; Morales et al., 2012; O'Neill et al., 2005; Roald et al., 2023; Street et al., 2025; Wang et al., 2009; Wong & Fuller, 2007). These approaches derive energy and reserve prices from the dual variables of power balance constraints, ensuring that prices reflect the marginal costs of maintaining security. However, most frameworks assume ancillary services costs are fully borne by consumers. Additionally, existing models fail to incorporate transmission revenues into reserve pricing, i.e., prices are derived without properly considering the transmission capacity required to deliver reserves, potentially leading to settlement imbalance. Furthermore, the nonconvex operational characteristics of generating units introduce additional distortions, often resulting in payments in addition to those resulting from the marginal prices, and several studies have sought to mitigate these inefficiencies (Ahunbay et al., 2024; Byers & Hug, 2023; O'Neill et al., 2017, 2005).

Early discussions on reserve cost allocation focused on generator outages. Proportional cost allocation based on generating units' capacity and unavailability was proposed in Strbac and Kirschen (2000). In Kirby and Hirst (2003), this idea was expanded with mechanisms incorporating historical outages and dispatch size. However, using historical outages as a proxy for predicting future events is suboptimal due to their low probability of occurrence, leading to erratic charges and misalignment between prospective reserve procurement and retrospective cost allocation. Specifically, if no outages occur, reserve costs lack clear entities to charge (Badesa et al., 2023). More recent studies, such as Liu et al. (2023) and Xiang et al. (2023), emphasize renewables and consumption variabilities as factors influencing reserve costs, relying on historical data. However, these variabilities are not the dominant driver of reserve needs in low-inertia systems (Matamala et al., 2024), and historical data often fail to predict future contingencies, underscoring the need for adaptive, causation-based allocation methods.

In Matamala et al. (2024), the costs of ORs in low-inertia systems are shown to be primarily driven by the generators with the largest capacity. The study evaluates proportional, Shapley value, and nucleolus methods, identifying the nucleolus as the one that prevents cross-subsidies, thereby promoting cooperative behavior. Building on this finding, the cost-causation principle is advocated in Badesa et al. (2025) to ensure reserve costs are borne by those creating the need for ancillary services. Furthermore, in Badesa et al. (2025), the authors highlight the need for cost-reflective frameworks that incentivize responsible behavior and investment in future grids. However, to date, no study combines a unified pricing model for energy, reserves, and transmission with a causation-based cost allocation methodology, leaving an important gap in the literature.

Within the context of an ISO-based market design, this paper addresses these issues by introducing a unified framework for the pricing and cost allocation of energy, reserves, and transmission services, grounded in causation-based principles. More specifically, the objective of this work is to extend the findings of Arroyo and Galiana (2005) to incorporate the cost-causation principle. To the best of the authors' knowledge, Arroyo and Galiana (2005) can be regarded as a foundational benchmark in the literature, as it is the first work to endogenously derive nodal marginal prices for both energy and reserves within a contingency-constrained co-optimization framework. This formulation has since underpinned a substantial body of subsequent research (Bouffard et al., 2005; Galiana et al., 2005; Karangelos & Bouffard, 2012; Wong & Fuller, 2007) and has also motivated practical developments in ISO-based market implementations, particularly in addressing reserve deliverability and post-contingency constraints (Chen et al., 2014; Wang & Chen, 2021).

Under a contingency-constrained setting, system operation is explicitly characterized for the pre-contingency state and for every contingency within the prescribed security criterion. Therefore, security is ensured as power balance is met without violating any technical limits for all system states under consideration. The adoption of such a setting is justified by the fact that the cost of secure operation is significantly lower than the cost of large-scale blackouts. The use of a contingency-constrained approach is consistent with established $N-k$ security standards, such as those of the North American Electric Reliability Corporation (NERC), which require system operators to restore a secure state immediately after a contingency, reflecting the critical nature of power grid infrastructure (North American Electric Reliability Corporation (NERC), 2026). These principles are also reflected in the operational procedures routinely implemented by system operators worldwide (Li & Hedman, 2019), including those of California (California ISO, 2025) and New York (New York ISO, 2026). It should be noted that the choice of the security standard is ultimately a regulatory decision, which directly shapes the cost-causation principle embedded in the market design.

The proposed adoption of the cost-causation principle prevents settlement imbalance, which is a major flaw of existing pricing methods even in the absence of nonconvexities in the market clearing. Such imbalances are here precluded by tying security charges to generating units that tighten security constraints. The security charge derived in this work can be understood as a causation-based remuneration adjustment to reflect the marginal contribution of each generator's outage to the total reserve cost. As a result, the proposed approach gives rise to a pricing system that aligns payments from consumers, which are the beneficiaries of security, with generation and transmission revenues while ensuring cost recovery for market participants. Moreover, this correction provides a sound basis for extensions to nonconvex market-clearing formulations and the analysis of market power abuse. We therefore contribute to knowledge with the following concepts:

1. A security charge that complements the prices of energy and reserves provided by the benchmark scheme proposed in Arroyo and Galiana (2005) so that reserve costs are efficiently allocated to the entities responsible for causing them,
2. the differentiation between up- and down-spinning reserve prices based on system's opportunity costs, and
3. a pricing system that properly remunerates transmission lines for the capacity set aside for contingencies that is used to ensure reserve deliverability through the network.

These newly proposed concepts rely on Lagrange multipliers, ensuring a transparent, fair, and efficient cost allocation under marginal pricing (Schweppe et al., 1988). Based on those concepts, we achieve a consistent settlement process that adheres to key axiomatic principles for energy and reserve contingency-constrained models. These principles include: (1) cost recovery, ensuring non-negative profit (surplus) for producers and consumers, and (2) system revenue adequacy, whereby total consumers' payment equals total generation and transmission revenue.

It is important to note that, while this study focuses on generation-driven cost causation, the same principles apply to demand-driven cost causation. Although frequency drops are the primary concern in most power systems, over-frequency events can be modeled analogously by considering sudden load disconnections instead of generation outages. In practice, such events are usually less critical than generator contingencies, but, with the advent of large data centers, load outages could become comparably significant.

The remainder of this paper is structured as follows. Section 2 develops the pricing framework for a simplified model without network constraints, considering only generator contingencies to place the focus on the concept of security charges and highlight their role in efficient market operations. In Section 3, the model is extended to incorporate network constraints, demand-side bidding, and both generator and line outages. This extension reinforces the necessity of security charges in more complex systems, examines their impact on the revenues of different market participants, including transmission assets, and introduces a novel approach for separately pricing up- and down-spinning reserves. Each section presents the corresponding optimization model for market clearing, introduces an example that is addressed using the benchmark pricing system, develops the proposed pricing framework, and provides and discusses the new results using the same example previously considered in the section. Section 4 reports our numerical experience with a larger case study based on the IEEE 300-bus system. Section 5 draws relevant conclusions and suggests directions for future research. Finally, the nomenclature is given in the Appendix.

2. Introducing the notion of security charges

This section introduces the concept of security charges under marginal pricing through a simplified contingency-constrained market-clearing model for energy and reserves. Briefly, the security charge concept is a cost allocation mechanism that reflects the different contributions of generators to the endogenously defined reserve needs, thereby addressing the limitations of existing uniform pricing in contingency-constrained models. Thus, higher charges are incurred by generators whose potential outages create higher reserve needs and costs, leading to a pricing framework that more accurately aligns costs with system's opportunity costs. This concept will be further developed in the following sections.

To that end, we first formulate the stylized optimization model used to introduce the security charge concept. Then, we revisit the existing uniform pricing framework in Arroyo and Galiana (2005), hereinafter referred to as the benchmark pricing scheme, and illustrate its limitations. Finally, we develop the notion of security charges based on the study of the Lagrangian dual function and how each term in this function should be attributed to each agent to ensure the cost-causation principle. The benefits of the proposed pricing system are illustrated through the assessment with the benchmark scheme.

2.1. Optimization model

For expository purposes, generation outages are considered using a contingency-constrained model (Wood et al., 2014), which can accommodate any instance of the practical deterministic $N-k$ security criterion (North American Electric Reliability Corporation (NERC), 2026). Moreover, a single-bus, single-period model is employed, considering an inelastic demand and linear offer cost functions, with a focus on spinning or synchronized reserves. For the sake of clarity in illustrating the cost-causation principle, this framework solely characterizes generation contingencies as full generator outages. Note, however, that the model could be extended to account for partial generator outages (derating). Such an extension is left for future work.

In this simplified framework, only one type of reserve, namely upward spinning reserves, is required to ensure system security. Thus, the market-clearing procedure is given by the following optimization problem, which is an instance of linear programming:

$$\min_{\boldsymbol{x}\geq\mathbf{0}} \quad \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \tag{1}$$

subject to:

$$\mathbb{1}^\top \boldsymbol{g}_0 = d \quad (\pi_0) \tag{2}$$

$$\mathbb{1}^\top \boldsymbol{g}_k = d \quad (\pi_k), \qquad \forall k \in \mathcal{K} \tag{3}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \quad \forall k \in \mathcal{K} \tag{4}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{5}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{6}$$

where $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$, $\mathbb{D}(\cdot)$ denotes the diagonal matrix operator, and $\mathbb{1}$ represents the all-ones vector. The pre-contingency state is denoted by subscript 0 while contingency states are indexed by $k \in \mathcal{K}$, with $\mathcal{K} = \{1, 2, \ldots, |\mathcal{K}|\}$.

The objective function to be minimized (1) consists of the sum of the costs for generating power and providing up-spinning reserves offered by the generators.

Constraints (2) and (3) ensure the power balance between generation and consumption under both pre-contingency and contingency states, respectively. Accordingly, the total output of all generators under every state must equal the load demand, which is assumed to be constant. Note that π_0 and π_k represent the associated Lagrange multipliers.

Constraint (4) relates the up-spinning reserve contributions to the power levels produced under the pre-contingency and contingency states through availability factors a_{ik}^g. Thus, for contingency states k where a given unit i is in service ($a_{ik}^g = 1$), the post-contingency output g_{ik} is bounded by the pre-contingency production plus the up-spinning reserve allocation. On the other hand, for contingency states k where the unit is out of service ($a_{ik}^g = 0$), the post-contingency production g_{ik} is forced to be zero. Lastly, constraints (5) and (6) set the operational limits of the generator outputs and reserves.

2.2. Challenges of the benchmark pricing framework

For contingency-constrained models such as problem (1)–(6), prices under a marginal pricing framework (Schweppe et al., 1988) can be computed using the benchmark scheme described in Arroyo and Galiana (2005), which is based on the use of Karush-Kuhn-Tucker (KKT) optimality conditions. Accordingly, the energy price is defined as the sum of the Lagrange multipliers corresponding to the power balance in both pre-contingency and contingency states. Moreover, reserves are priced through the so-called security price, which is defined as the sum of the Lagrange multipliers related to the power balance equations under contingency. As a result, for problem (1)–(6), generators collect revenues for their pre-contingency power outputs and scheduled up-spinning reserves using the aforementioned energy and security prices, respectively. Analogously, consumers are charged the energy price for

Table 1
Prices and settlement for problem (1)–(6) according to the benchmark pricing system.

Energy Price	$p^e = \pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*$
Security Price	$p^s = \sum_{k\in\mathcal{K}} \pi_k^*$
Generation Energy Revenue	$R_i^{g,e} = p^e g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^s r_i^{g,up}$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up}$
Consumer Payment	$CP = p^e d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^g = R_i^{g,t} - C_i^{g,t}$

Table 2
Single-bus example – Generation data.

i	$\overline{G}_i$ (MW)	$\overline{R}_i^{g,up}$ (MW)	c_i ($/MWh)	$q_i^{g,up}$ ($/MW/h)
1	100	50	20	2
2	60	30	50	5
3	70	35	100	10

Table 3
Single-bus example – Optimal results (MW).

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Generation under Contingency		
			Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
1	65	0	0	65	65
2	30	30	60	0	55
3	25	35	60	55	0
Total	120	65	120	120	120

Table 4
Single-bus example – Lagrange multipliers ($/MWh).

Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3
20	80	0	0

their consumption. Table 1 summarizes the prices and settlement for the market-clearing model (1)–(6) according to the benchmark pricing scheme (Arroyo & Galiana, 2005).

We now apply this benchmark pricing framework to an illustrative example involving three generators. Generation data are provided in Table 2. The load is 120 MW. Three contingencies are considered here, each defined by the outage of each generator. Using the simplex algorithm of CPLEX, problem (1)–(6) has been solved to optimality. Table 3 presents the optimal solution, which features a total cost equal to $5,800. As can be seen, the optimization prioritizes energy and reserve offers in reverse order, scheduling the least-cost generator (generator 1) exclusively to supply power in the pre-contingency state. In contrast, generators 2 and 3 are used for both power provision in the pre-contingency state and up-spinning reserve. Note that the up-spinning reserve contributions of generators 2 and 3, which are respectively equal to their corresponding upper bounds, amount to the pre-contingency power output of generator 1. It is also worth mentioning that the up-spinning reserve contributions of generators 2 and 3 exceed the reserve needed to guard against the loss of generators 3 and 2, respectively. In other words, the outage of generator 1 is the critical contingency state.

The Lagrange multipliers associated with the power balance equations in the pre-contingency and contingency states are listed in Table 4. For the outages of generators 2 and 3, the associated multipliers are zero, as the loss of these generators does not constrain the system. In contrast, the multiplier for the outage of generator 1 is different from zero as an infinitesimal perturbation of the corresponding power balance equation would yield a different solution, thereby resulting in a change in the value of the objective function.

Table 5
Single-bus example – Prices according to the benchmark pricing system ($/MWh).

Energy Price, p^e	Security Price, p^s
100	80

Table 6
Single-bus example – Generation settlement according to the benchmark pricing system ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	6,500	1,300	0	1,300	5,200
2	3,000	2,400	5,400	1,500	150	1,650	3,750
3	2,500	2,800	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**17,200**	**5,300**	**500**	**5,800**	**11,400**

Table 7
Single-bus example – System settlement according to the benchmark pricing system ($).

Generation Revenue	17,200
Consumer Payment	12,000
Balance	–5,200

Tables 5 and 6 respectively report the prices and settlement for the generators as per the definitions of Table 1. Also, according to Table 1, the consumer payment is given by $p^e d = 100 \times 120 = \$12,000$. Interestingly, the benchmark pricing scheme yields a total generation revenue exceeding the total consumer payment by $5,200, a distortion generated by the pricing system, as highlighted in Table 7. This mismatch is addressed by the novel approach described in the next section.

2.3. Proposed pricing system

As an alternative to the benchmark scheme, the proposed pricing system acknowledges the contribution of each generator to keep the power balance under each contingency state. To that end, we explore the structure of the Lagrangian dual (LD) problem to ensure a market with system revenue adequacy where the total generation revenue is equal to the consumer payment. When strong duality holds, which is guaranteed for feasible linear programs, the LD problem yields the optimal solution to the original problem. Moreover, the LD problem can be interpreted as a price-based coordination mechanism among market agents, where Lagrange multipliers play the role of prices (Conejo et al., 1999). For problem (1)–(6), the LD function is built as follows (Conejo et al., 1999):

$$\phi(\boldsymbol{\Pi}) = \min_{\boldsymbol{x}\in\mathcal{X}} \; \boldsymbol{c}^\top \boldsymbol{g}_0 + \left(\boldsymbol{q}^{g,up}\right)^\top \boldsymbol{r}^{g,up} + \pi_0 \left(d - \mathbb{1}^\top \boldsymbol{g}_0\right) + \sum_{k\in\mathcal{K}} \pi_k \left(d - \mathbb{1}^\top \boldsymbol{g}_k\right) \tag{7}$$

where $\boldsymbol{\Pi} = \{\pi_0, \{\pi_k\}_{k\in\mathcal{K}}\}$ and $\mathcal{X}$ includes constraints (4)–(6) and $\boldsymbol{x} \geq \boldsymbol{0}$.

Taking the terms that do not depend on $\boldsymbol{x}$ out of the minimization problem and factoring out yields:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d + \min_{\boldsymbol{x}\in\mathcal{X}} \left\{ \left(\boldsymbol{c}^\top - \pi_0 \mathbb{1}^\top\right) \boldsymbol{g}_0 + \left(\boldsymbol{q}^{g,up}\right)^\top \boldsymbol{r}^{g,up} - \sum_{k\in\mathcal{K}} \pi_k \mathbb{1}^\top \boldsymbol{g}_k \right\} \tag{8}$$

The minimization problem in (8) is generator-wise separable, as no coupling constraints are present, i.e., there are no system-wide constraints that link actions across all generators. This separability reveals the incentive compatibility of the proposed pricing framework, as each generator maximizes its individual profit at the optimal solution.

Besides, by factoring out the negative sign in (8), we can transform the *min* operator into a *max* operator. Thus, rewriting Eq. (8) accordingly, we have Eq. (9) as follows:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) \right] \tag{9}$$

where $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}\}$, $\mathcal{X}_i$ is the subset of constraints related to $\boldsymbol{x}_i$, i.e., constraints (5), (6), and $\boldsymbol{x}_i \geq \mathbf{0}$, whereas $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ stands for the revenue fraction of generator i under contingency k due to its best response against post-contingency Lagrange multiplier π_k, given its pre-contingency dispatch and reserve schedule, $\boldsymbol{x}_i$, and availability status at state k. Note that $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ can be cast as:

$$\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \quad \pi_k g_{ik} \tag{10}$$

subject to:

$$g_{ik} \leq a_{ik}^{g} \left(g_{i0} + r_i^{g,up}\right) \tag{11}$$

Based on KKT conditions, π_k can only take non-negative values at the optimal solution to (1)–(6). As a consequence, we have two possible outputs for $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$: (1) If π_k is positive, g_{ik}^{*} will take its maximum possible value, given by $g_{ik}^{*} = a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$, and $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will be equal to $\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$; (2) If π_k is zero, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will be 0. Thus, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ will only take values different from zero when π_k is positive. Hence, $\psi_{ik}^{g}(\pi_k, \boldsymbol{x}_i)$ can be equivalently replaced with $\pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right)$. Rewriting (9) accordingly, we have Eq. (12) as follows:

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g}\left(g_{i0} + r_i^{g,up}\right) \right] \tag{12}$$

By rearranging Eq. (12), we obtain Eq. (13):

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k a_{ik}^{g} - q_i^{g,up}\right) r_i^{g,up} \right] \tag{13}$$

The first term in the right-hand side of (13) represents the consumer payment. The terms in square brackets represent the profit of generator i, which is made up of the net revenue, i.e., revenue minus cost, from both selling energy and providing up-spinning reserve. In conventional models, all generators are paid uniform prices for energy and reserves. However, when contingencies are explicitly modeled, cost causation also becomes explicit in the final optimal reserve allocation cost. In this case, generators receive no remuneration in the contingency states where they fail to be available to contribute to energy and reserves. This is reflected in the second and third terms of expression (13), where the binary availability parameter a_{ik}^{g} nullifies the contribution of the Lagrange multipliers in the contingency states where the generator is not available. Nevertheless, uniform pricing does not account for availability parameters. Therefore, rewriting (13) accordingly, we have expression (14):

$$\phi(\boldsymbol{\Pi}) = \left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k\right) d - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left[\left(\pi_0 + \sum_{k\in\mathcal{K}} \pi_k - c_i\right) g_{i0} + \left(\sum_{k\in\mathcal{K}} \pi_k - q_i^{g,up}\right) r_i^{g,up} - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k \left(g_{i0} + r_i^{g,up}\right) \right] \tag{14}$$

Table 8
Proposed pricing system and settlement for problem (1)–(6).

Energy Price	$p^{e} = \pi_0^{*} + \sum_{k\in\mathcal{K}} \pi_k^{*}$
Up-Spinning Reserve Price	$p^{up} = \sum_{k\in\mathcal{K}} \pi_k^{*}$
Generation Energy Revenue	$R_i^{g,e} = p^{e} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R_i^{g,up} = p^{up} r_i^{g,up}$
Generation Security Charge	$C_i^{g,s} = \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^{*}\left(g_{i0} + r_i^{g,up}\right)$
Generation Total Revenue	$R_i^{g,t} = R_i^{g,e} + R_i^{g,up} - C_i^{g,s}$
Consumer Payment	$CP = p^{e} d$
Generation Energy Cost	$C_i^{g,e} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C_i^{g,up} = q_i^{g,up} r_i^{g,up}$
Generation Total Cost	$C_i^{g,t} = C_i^{g,e} + C_i^{g,up}$
Generation Profit	$Profit_i^{g} = R_i^{g,t} - C_i^{g,t}$

where the set $\mathcal{K}_i^{OFF}$ explicitly identifies the contingency states in which generator i is out of service. The first term within the square brackets represents the net revenue of generator i from selling energy. Note that, at the optimal solution, energy is priced at $\pi_0^{*} + \sum_{k\in\mathcal{K}} \pi_k^{*}$, which is identical to the price charged to consumers (first term in the right-hand side of (14)). Moreover, this price is identical to that derived in the benchmark pricing system (Table 1). The second term within the square brackets represents the net revenue from up-spinning reserve, where such a commodity is priced at $\sum_{k\in\mathcal{K}} \pi_k^{*}$, which is consistent with the security price defined in the benchmark pricing system (Table 1). The last term in (14) is here coined as "security charge" and can be viewed as what generator i should pay back for being responsible for system required reserves, which is aligned with the cost-causation principle. In other words, this term can be interpreted as the payback of the portion of the security revenue not secured by this generator, i.e., the part involving Lagrange multipliers associated with an availability parameter equal to 0. Importantly, while energy and reserve prices are uniform system-wide, the security charge is generator-specific.

Therefore, different from the settlement associated with the benchmark pricing system, the total revenue for each generator consists of three components: the energy revenue (energy price times pre-contingency power output), the up-spinning reserve revenue (up-spinning reserve price times up-spinning reserve), and the security charge (sum of the corresponding post-contingency Lagrange multipliers times the sum of pre-contingency power output and up-spinning reserve). Table 8 shows the proposed pricing scheme and the corresponding settlement.

The proposed pricing system has been applied to the illustrative example described in Section 2.2. Using the results provided in Tables 3 and 4 in the expressions listed in Table 8 gives rise to the results reported in Table 9, which lists the breakdown of revenues, costs, charges, and profits for the three generators. Compared to the results reported in Table 6, the revenues from energy and reserves and the associated costs are identical. However, as a major salient result, generator 1 is subject to a security charge under the proposed pricing scheme. Note that this generator is the primary driver of the reserve requirement, as indicated by the non-zero Lagrange multiplier in Table 4. This charge accounts for the increased system costs associated with the need for up-spinning reserves. Specifically, generator 1's failure requires 65 MW of up-spinning reserve. As a result, under the proposed method, the total generation revenue is $12,000, which exactly matches the consumer payment (Table 10), thereby overcoming the issue featured by the benchmark approach. As can be seen in Table 9, the consideration of the security charge reduces the profit for generator 1 compared to that reported in Table 6. Note, however, that the new resulting profit is non-negative.

The following theorems are stated for the proposed pricing framework, with their respective proofs provided subsequently.

Table 9
Single-bus example – Proposed generation settlement ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,t}$	$Profit_i^g$
1	6,500	0	5,200	1,300	1,300	0	1,300	0
2	3,000	2,400	0	5,400	1,500	150	1,650	3,750
3	2,500	2,800	0	5,300	2,500	350	2,850	2,450
Total	**12,000**	**5,200**	**5,200**	**12,000**	**5,300**	**500**	**5,800**	**6,200**

Table 10
Single-bus example – Proposed system settlement ($).

Generation Revenue	12,000
Consumer Payment	12,000
Balance	0

Theorem 1 (*Cost Recovery*)**.** *The proposed pricing framework ensures that generators' profits are non-negative.*

Proof. Let us rewrite the LD function in Eq. (13) as follows:

$$\phi(\boldsymbol{\Pi}) = CP(\boldsymbol{\Pi}) - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \left\{ Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i) \right\} \tag{15}$$

where $CP(\boldsymbol{\Pi})$ and $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ represent the consumer payment and the profit of generator i, respectively. Since $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ is a linear objective function maximized over $\boldsymbol{x}_i$, with $\boldsymbol{x}_i \geq \mathbf{0}$, the optimal value of $Profit_i^g(\boldsymbol{\Pi}, \boldsymbol{x}_i)$ is guaranteed to be non-negative. In other words, if π_0 and π_k fail to cover generator i's costs, the generator will not produce any output, resulting in $g_{i0}^* = r_i^{g,up*} = 0$. Likewise, at the optimal solution $\boldsymbol{x}_i^*$ and $\boldsymbol{\Pi}^*$:

$$Profit_i^g(\boldsymbol{\Pi}^*, \boldsymbol{x}_i^*) \geq 0, \ \forall i \in \mathcal{I} \tag{16}$$

Therefore, the theorem holds. □

Theorem 2 (*System Revenue Adequacy*)**.** *The total consumer payment is equal to the total generation revenue.*

Proof. By the strong duality theorem, the optimal value of the primal objective function in (1) is equal to the optimal value of the LD function in Eq. (14). Thus, at the optimal solution $\boldsymbol{x}^*$ and $\boldsymbol{\Pi}^*$, we have:

$$\begin{aligned}\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} = {} & \left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d \\ & - \sum_{i\in\mathcal{I}} \left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^* - c_i\right) g_{i0}^* \right. \\ & + \left(\sum_{k\in\mathcal{K}} \pi_k^* - q_i^{g,up}\right) r_i^{g,up*} \\ & \left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{17}$$

By canceling out identical terms in both sides of Eq. (17) and rearranging terms, we simplify the expression as follows:

$$\begin{aligned}\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) d = \sum_{i\in\mathcal{I}} & \left[\left(\pi_0^* + \sum_{k\in\mathcal{K}} \pi_k^*\right) g_{i0}^* + \sum_{k\in\mathcal{K}} \pi_k^* r_i^{g,up*} \right. \\ & \left. - \sum_{k\in\mathcal{K}_i^{OFF}} \pi_k^* \left(g_{i0}^* + r_i^{g,up*}\right) \right]\end{aligned} \tag{18}$$

Therefore, at the optimal solution, the consumer payment equals the sum of all generator revenues. □

3. Separately pricing up and down reserves and transmission pricing

In this section, the scope is broadened to consider a more realistic setting. To that end, network constraints and demand-side bidding are now incorporated into the market-clearing model, whereas both generation and line outages are accounted for. As a consequence, the set of ancillary services is extended to also include downward reserves, which are co-optimized with energy and upward reserves. This model is useful to comprehensively present the salient features of the proposed pricing scheme compared to the benchmark pricing system, which, aside from the concept of security charges already discussed in Section 2, comprise (1) separately pricing upward and downward reserves, and (2) transmission pricing.

3.1. Optimization model

The market-clearing model is formulated as the following linear program:

$$\begin{aligned}\min_{\substack{x\geq 0, y\geq 0,\\ \theta_0, \theta_k}} \quad & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} - \boldsymbol{w}^\top \boldsymbol{d}_0 \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\end{aligned} \tag{19}$$

subject to:

$$\boldsymbol{M}^g \boldsymbol{g}_0 + \boldsymbol{A}\boldsymbol{H}\boldsymbol{\theta}_0 = \boldsymbol{M}^d \boldsymbol{d}_0 \quad (\boldsymbol{\pi_0}) \tag{20}$$

$$\boldsymbol{M}^g \boldsymbol{g}_k + \boldsymbol{A}_k \boldsymbol{H}_k \boldsymbol{\theta}_k = \boldsymbol{M}^d \boldsymbol{d}_k \quad (\boldsymbol{\pi}_k), \qquad \forall k \in \mathcal{K} \tag{21}$$

$$-\boldsymbol{F}^{ss} \leq \boldsymbol{H}\boldsymbol{\theta}_0 \leq \boldsymbol{F}^{ss} \quad \left(\boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}\right) \tag{22}$$

$$-\boldsymbol{F}^{st} \leq \boldsymbol{H}_k \boldsymbol{\theta}_k \leq \boldsymbol{F}^{st} \quad \left(\boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right), \qquad \forall k \in \mathcal{K} \tag{23}$$

$$\boldsymbol{g}_k \leq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 + \boldsymbol{r}^{g,up}), \qquad \forall k \in \mathcal{K} \tag{24}$$

$$\boldsymbol{g}_k \geq \mathbb{D}(\boldsymbol{a}_k^g)(\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn}), \qquad \forall k \in \mathcal{K} \tag{25}$$

$$\boldsymbol{g}_0 + \boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{G}} \tag{26}$$

$$\boldsymbol{g}_0 - \boldsymbol{r}^{g,dn} \geq \mathbf{0} \tag{27}$$

$$\boldsymbol{r}^{g,up} \leq \overline{\boldsymbol{R}}^{g,up} \tag{28}$$

$$\boldsymbol{r}^{g,dn} \leq \overline{\boldsymbol{R}}^{g,dn} \tag{29}$$

$$\boldsymbol{d}_k \geq \boldsymbol{d}_0 - \boldsymbol{r}^{d,up}, \qquad \forall k \in \mathcal{K} \tag{30}$$

$$\boldsymbol{d}_k \leq \boldsymbol{d}_0 + \boldsymbol{r}^{d,dn}, \qquad \forall k \in \mathcal{K} \tag{31}$$

$$\boldsymbol{d}_0 - \boldsymbol{r}^{d,up} \geq \mathbf{0} \tag{32}$$

$$\boldsymbol{d}_0 + \boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{D}} \tag{33}$$

$$\boldsymbol{r}^{d,up} \leq \overline{\boldsymbol{R}}^{d,up} \tag{34}$$

$$\boldsymbol{r}^{d,dn} \leq \overline{\boldsymbol{R}}^{d,dn} \tag{35}$$

where $\boldsymbol{x} = \{\boldsymbol{g}_0, \boldsymbol{r}^{g,up}, \boldsymbol{r}^{g,dn}, \{\boldsymbol{g}_k\}_{k\in\mathcal{K}}\}$, $\boldsymbol{y} = \{\boldsymbol{d}_0, \boldsymbol{r}^{d,up}, \boldsymbol{r}^{d,dn}, \{\boldsymbol{d}_k\}_{k\in\mathcal{K}}\}$, $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta}_k$ are the unbounded nodal phase angles, $\boldsymbol{H} = \boldsymbol{X}^{-1}\boldsymbol{A}^\top$, and $\boldsymbol{H}_k = \boldsymbol{X}^{-1}\boldsymbol{A}_k^\top$. Note also that relevant Lagrange multipliers are shown in parentheses.

As per (19), the optimization goal is the minimization of the sum of the costs for generating power and providing up- and down-spinning reserves offered by the generators minus the sum of the bid utility functions for consuming power plus the costs for providing up- and down-spinning reserves by the loads. This optimization goal is equivalent to maximizing the social welfare.

As done in Arroyo and Galiana (2005), and for presentation convenience, a dc power flow model relying on nodal phase angles (Wood et al., 2014) is used. Note, however, that the alternative dc power flow formulation based on sensitivity factors, which is typically adopted in industry practice, can be equivalently used. Constraints (20) and (21) represent the nodal power balance equations under the pre-contingency and contingency states, respectively. Constraints (22) and (23) enforce the line flow capacity limits, thereby explicitly bounding the differences of phase angles of connected buses. Additionally, to represent a possible flow-relief adjustment under stressed operating conditions, a flow-relief vector $\boldsymbol{\gamma}$ is introduced to relate short-term and steady-state line ratings through the entry-wise product $\boldsymbol{\gamma}\boldsymbol{F}^{ss}$, i.e., $\boldsymbol{F}^{st} = \boldsymbol{\gamma}\boldsymbol{F}^{ss}$. Line outages

Table 11
Nodal prices for problem (19)–(35) according to the benchmark pricing system.

Nodal Energy Price	$p^e_b = \pi^*_{b0} + \sum_{k\in\mathcal{K}} \pi^*_{bk}$
Nodal Security Price	$p^s_b = \sum_{k\in\mathcal{K}} \pi^*_{bk}$

Table 12
Generation settlement for problem (19)–(35) according to the benchmark pricing system.

Generation Energy Revenue	$R^{g,e}_i = p^e_{b(i)} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R^{g,up}_i = p^s_{b(i)} r^{g,up}_i$
Generation Down-Spinning Reserve Revenue	$R^{g,dn}_i = p^s_{b(i)} r^{g,dn}_i$
Generation Total Revenue	$R^{g,t}_i = R^{g,e}_i + R^{g,up}_i + R^{g,dn}_i$
Generation Energy Cost	$C^{g,e}_i = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C^{g,up}_i = q^{g,up}_i r^{g,up}_i$
Generation Down-Spinning Reserve Cost	$C^{g,dn}_i = q^{g,dn}_i r^{g,dn}_i$
Generation Total Cost	$C^{g,t}_i = C^{g,e}_i + C^{g,up}_i + C^{g,dn}_i$
Generation Profit	$Profit^g_i = R^{g,t}_i - C^{g,t}_i$

are incorporated by explicitly modifying the network topology for each contingency k. This is achieved by constructing a contingency-specific incidence matrix, $\boldsymbol{A_k}$, obtained from the base-case incidence matrix $\boldsymbol{A}$ by setting to zero the column associated with the out-of-service line. The modified matrix $\boldsymbol{A_k}$ ensures that the disconnected line carries no flow and does not contribute to the nodal power balance under that contingency. In such a case, the flow-mapping matrix $\boldsymbol{H_k}$, which relates bus voltage angles to line flows, is updated accordingly. The superscripts "+" and "−" on the Lagrange multipliers for (22) and (23) indicate their association with the left- and right-hand inequalities, respectively. According to KKT conditions, the Lagrange multipliers for the left inequalities ($\boldsymbol{\pi}^{f+}_0, \boldsymbol{\pi}^{f+}_k$) are non-negative, while those for the right inequalities ($\boldsymbol{\pi}^{f-}_0, \boldsymbol{\pi}^{f-}_k$) are non-positive.

Admittedly, the precise characterization of the effect of the transmission network should rely on an ac power flow model, especially for systems under stress (Frank & Rebennack, 2016). Notwithstanding, the dc model is the standard network characterization in market design and pricing practice worldwide, including state-of-the-art contingency-constrained models such as that presented in Arroyo and Galiana (2005), upon which this work is built.

Constraints (24) and (25) relate the up- and down-spinning reserve contributions from generators to the production levels under the pre-contingency and contingency states.

Furthermore, constraints (26)–(29) ensure that such generation and reserve levels remain within their respective operational boundaries.

Constraints (30) and (31) establish the relationship between the up- and down-spinning reserve contributions from consumers and the power levels consumed under the pre-contingency and contingency states. Finally, demand-related bounds are set in constraints (32)–(35).

3.2. Challenges of the benchmark pricing framework

According to Arroyo and Galiana (2005), nodal prices for energy and security can be defined for problem (19)–(35), as summarized in Table 11. The corresponding generation and consumer settlements are provided in Tables 12 and 13, respectively. As can be seen, for each bus, a single price, namely the nodal security price, is used for all spinning reserves provided by the generators and consumers at that bus.

For illustration purposes, the benchmark pricing system presented in Table 11 is applied to the two-bus, one-line system depicted in Fig. 1, where the generation fleet is based on that considered in Section 2.2. The line reactance is 1 p.u. on a base of 100 MVA and 138 kV, whereas the steady-state and short-term line capacities are both 70 MW, i.e., the

Table 13
Consumer settlement for problem (19)–(35) according to the benchmark pricing system.

Consumer Payment for Energy	$CP^e_j = p^e_{b(j)} d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R^{d,up}_j = p^s_{b(j)} r^{d,up}_j$
Consumer Down-Spinning Reserve Revenue	$R^{d,dn}_j = p^s_{b(j)} r^{d,dn}_j$
Consumer Utility	$U^d_j = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C^{d,up}_j = q^{d,up}_j r^{d,up}_j$
Consumer Down-Spinning Reserve Cost	$C^{d,dn}_j = q^{d,dn}_j r^{d,dn}_j$
Consumer Total Cost	$C^{d,t}_j = C^{d,up}_j + C^{d,dn}_j$
Consumer Payment	$CP_j = CP^e_j - R^{d,up}_j - R^{d,dn}_j$
Consumer Profit	$Profit^d_j = U^d_j - C^{d,t}_j - CP_j$

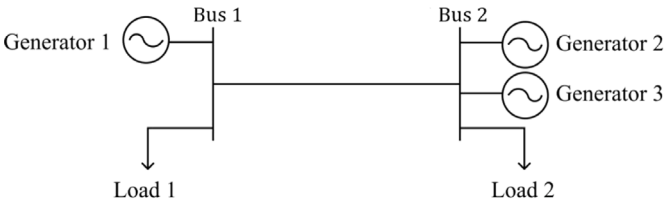


Fig. 1. Two-bus example – One-line diagram.

Table 14
Two-bus example – Generation data.

i	$\overline{G}_i$ (MW)	$\overline{R}^{g,up}_i$ (MW)	$\overline{R}^{g,dn}_i$ (MW)	c_i ($/MWh)	$q^{g,up}_i$ ($/MW/h)	$q^{g,dn}_i$ ($/MW/h)
1	100	50	50	20	2	2
2	60	30	30	50	5	5
3	70	35	35	100	10	10

Table 15
Two-bus example – Load data.

j	$\overline{D}_j$ (MW)	$\overline{R}^{d,up}_j$ (MW)	$\overline{R}^{d,dn}_j$ (MW)	w_j ($/MWh)	$q^{d,up}_j$ ($/MW/h)	$q^{d,dn}_j$ ($/MW/h)
1	90	10	10	200	150	300
2	40	10	10	150	100	250

corresponding flow-relief parameter is equal to 1. In Table 14, the generation data presented in Table 2 are extended with the information for down-spinning reserve offers, which is the same as for up-spinning reserves. Load data are provided in Table 15. The analysis considers four contingencies: the failure of each generator and the outage of the line.

Problem (19)–(35) has been solved to optimality using the simplex algorithm of CPLEX. Table 16 displays the optimal results, which feature a total value of the objective function equal to −$15,475. In other words, the optimal social welfare amounts to $15,475. It should be noted that the optimal pre-contingency system demand amounts to 120 MW, which is identical to the value of the inelastic demand in the single-bus example. Consequently, the optimal generation schedule is similar to that attained for the single-bus example, the main difference being the 5 MW of down-spinning reserve scheduled to generator 2. It is also worth highlighting that load 1 is scheduled to provide upward spinning reserve, which is deployed in response to contingencies involving the loss of generator 1 or line 1–2.

The outage of generator 1 constitutes a critical contingency for the system, requiring the full deployment of the up-spinning reserves scheduled to generators 2 and 3 and load 1. Analogously, the line outage is another critical contingency that compels generator 2 to reduce its output by fully utilizing its down-spinning reserve. Thus, for

Table 16
Two-bus example – Optimal results (MW).

Generator	Pre-Contingency Generation	Up-Spinning Reserve	Down-Spinning Reserve	Generation under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	75	0	0	0	75	75	75
2	30	30	5	60	0	45	25
3	15	35	0	50	45	0	15
Total	**120**	**65**	**5**	**110**	**120**	**120**	**115**
Load	Pre-Contingency Demand	Up-Spinning Reserve	Down-Spinning Reserve	Demand under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	80	10	0	70	80	80	75
2	40	0	0	40	40	40	40
Total	**120**	**10**	**0**	**110**	**120**	**120**	**115**
Line	Pre-Contingency Power Flow 1–2			Power Flow 1–2 under Contingency			
				Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1–2	−5			−70	−5	−5	0

Table 17
Two-bus example – Lagrange multipliers for the nodal power balance equations ($/MWh).

Bus	Pre-Contingency State	Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2
1	20	180	0	0	0
2	20	85	0	0	−5

this particular example, the line outage is the sole reason for downward reserves. Note, however, that under this contingency state a marginal increase in load 2 would actually reduce the need for down-spinning reserves at this bus. This explains why the corresponding Lagrange multiplier is negative (Table 17). Finally, it is worth pointing out that line congestion is solely experienced during the outage of generator 1. In fact, the pre-contingency power flow is well below the line capacity to allow generators 2 and 3 to ramp up during this critical outage.

The nodal prices and the settlement for generators and consumers calculated according to Tables 11, 12, and 13 are presented in Tables 18, 19, and 20, respectively. As can be seen, the profit distribution among market agents is economically inconsistent as the total profit of generators and consumers is 44.6% greater than the optimal social welfare. Furthermore, similar to the previous example, the benchmark pricing scheme results in total generation revenue exceeding total consumer payment, as summarized in Table 21. This issue becomes more important in a network-constrained setting because the absence of a transmission surplus to compensate for lines creates an additional financial burden. For this particular example, the consumer payment is insufficient to cover the generation revenue and the transmission revenue associated with the line congestion under the outage of generator 1. The resulting settlement imbalance may be addressed by a compensation in addition to the remuneration from the marginal prices that may give rise to discrimination among market participants and inadequate economic signals, among other issues. Alternatively, in Section 3.3 we propose a new pricing scheme that does not feature settlement imbalance while relying on a sound mathematical framework.

3.3. Proposed pricing system

Following the procedure described in Section 2.3, we start by building the LD function through the dualization of constraints (20)–(23),

Table 18
Two-bus example – Nodal prices according to the benchmark pricing system ($/MWh).

b	Nodal Energy Price, p^e_b	Nodal Security Price, p^s_b
1	200	180
2	100	80

Table 19
Two-bus example – Generation settlement according to the benchmark pricing system ($).

i	$R^{g,e}_i$	$R^{g,up}_i$	$R^{g,dn}_i$	$R^{g,t}_i$	$C^{g,e}_i$	$C^{g,up}_i$	$C^{g,dn}_i$	$C^{g,t}_i$	$Profit^g_i$
1	15,000	0	0	15,000	1,500	0	0	1,500	13,500
2	3,000	2,400	400	5,800	1,500	150	25	1,675	4,125
3	1,500	2,800	0	4,300	1,500	350	0	1,850	2,450
Total	**19,500**	**5,200**	**400**	**25,100**	**4,500**	**500**	**25**	**5,025**	**20,075**

Table 20
Two-bus example – Consumer settlement according to the benchmark pricing system ($).

j	CP^e_j	$R^{d,up}_j$	$R^{d,dn}_j$	CP_j	U^d_j	$C^{d,up}_j$	$C^{d,dn}_j$	$C^{d,t}_j$	$Profit^d_j$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

Table 21
Two-bus example – System settlement according to the benchmark pricing system ($).

Generation Revenue	25,100
Consumer Payment	18,200
Balance	−6,900

giving rise to Eq. (36):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \boldsymbol{\theta_0},\boldsymbol{\theta_k}}} \Bigg\{ & \boldsymbol{c}^\top \boldsymbol{g}_0 + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} \\ & - \boldsymbol{w}^\top \boldsymbol{d}_0 + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & + \boldsymbol{\pi}_0^\top \left(\boldsymbol{M}^d \boldsymbol{d}_0 - \boldsymbol{M}^g \boldsymbol{g}_0 - \boldsymbol{A}\boldsymbol{H}\boldsymbol{\theta}_0\right) \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \left(\boldsymbol{M}^d \boldsymbol{d}_k - \boldsymbol{M}^g \boldsymbol{g}_k - \boldsymbol{A}_k \boldsymbol{H}_k \boldsymbol{\theta}_k\right)\right] \\ & + \left(\boldsymbol{\pi}_0^{f+}\right)^\top \left(-\boldsymbol{F}^{ss} - \boldsymbol{H}\boldsymbol{\theta}_0\right) + \left(\boldsymbol{\pi}_0^{f-}\right)^\top \left(\boldsymbol{F}^{ss} - \boldsymbol{H}\boldsymbol{\theta}_0\right) \\ & + \sum_{k\in\mathcal{K}} \Bigg[\left(\boldsymbol{\pi}_k^{f+}\right)^\top \left(-\boldsymbol{F}^{st} - \boldsymbol{H}_k\, \boldsymbol{\theta}_k\right) \\ & + \left(\boldsymbol{\pi}_k^{f-}\right)^\top \left(\boldsymbol{F}^{st} - \boldsymbol{H}_k \boldsymbol{\theta}_k\right) \Bigg] \Bigg\} \end{aligned} \tag{36}$$

where $\boldsymbol{\Pi} = \left\{\boldsymbol{\pi}_0, \boldsymbol{\pi}_0^{f+}, \boldsymbol{\pi}_0^{f-}, \left\{\boldsymbol{\pi}_k, \boldsymbol{\pi}_k^{f+}, \boldsymbol{\pi}_k^{f-}\right\}_{k\in\mathcal{K}}\right\}$, $\mathcal{X}$ represents the feasibility set associated with generation-related constraints $\boldsymbol{x} \geq \boldsymbol{0}$ and (24)–(29), whereas $\mathcal{Y}$ is the feasibility set corresponding to consumer-related constraints $\boldsymbol{y} \geq \boldsymbol{0}$ and (30)–(35). Rearranging (36), we have:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = \min_{\substack{\boldsymbol{x}\in\mathcal{X},\\ \boldsymbol{y}\in\mathcal{Y},\\ \boldsymbol{\theta_0},\boldsymbol{\theta_k}}} \Bigg\{ & \left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn} + \left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 \\ & + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn} \\ & - \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + (\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-})^\top\right] \boldsymbol{H}\boldsymbol{\theta}_0 \\ & - \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + (\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-})^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \\ & - \left[(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-})^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} (\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-})^\top \boldsymbol{F}^{st}\right] \Bigg\} \end{aligned} \tag{37}$$

The last square-bracketed term in Eq. (37) is independent of $\boldsymbol{x}$ and $\boldsymbol{y}$, allowing it to be factored out of the minimization problem. Moreover, leveraging the separability of the resulting minimization problem, Eq. (37) can be equivalently rewritten as Eq. (38):

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\left[\left(\boldsymbol{\pi}_0^{f+} - \boldsymbol{\pi}_0^{f-}\right)^\top \boldsymbol{F}^{ss} + \sum_{k\in\mathcal{K}} \left(\boldsymbol{\pi}_k^{f+} - \boldsymbol{\pi}_k^{f-}\right)^\top \boldsymbol{F}^{st}\right] \\ & + \min_{\boldsymbol{x}\in\mathcal{X}} \Big[\left(\boldsymbol{c}^\top - \boldsymbol{\pi}_0^\top \boldsymbol{M}^g\right) \boldsymbol{g}_0 - \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^g \boldsymbol{g}_k + (\boldsymbol{q}^{g,up})^\top \boldsymbol{r}^{g,up} \\ & + (\boldsymbol{q}^{g,dn})^\top \boldsymbol{r}^{g,dn}\Big] + \min_{\boldsymbol{y}\in\mathcal{Y}} \Big[\left(\boldsymbol{\pi}_0^\top \boldsymbol{M}^d - \boldsymbol{w}^\top\right) \boldsymbol{d}_0 + \sum_{k\in\mathcal{K}} \boldsymbol{\pi}_k^\top \boldsymbol{M}^d \boldsymbol{d}_k \\ & + (\boldsymbol{q}^{d,up})^\top \boldsymbol{r}^{d,up} + (\boldsymbol{q}^{d,dn})^\top \boldsymbol{r}^{d,dn}\Big] \\ & - \min_{\boldsymbol{\theta_0},\boldsymbol{\theta_k}} \Bigg\{ \left[\boldsymbol{\pi}_0^\top \boldsymbol{A} + \left(\boldsymbol{\pi}_0^{f+} + \boldsymbol{\pi}_0^{f-}\right)^\top\right] \boldsymbol{H}\boldsymbol{\theta}_0 \\ & + \sum_{k\in\mathcal{K}} \left[\boldsymbol{\pi}_k^\top \boldsymbol{A}_k + \left(\boldsymbol{\pi}_k^{f+} + \boldsymbol{\pi}_k^{f-}\right)^\top\right] \boldsymbol{H}_k \boldsymbol{\theta}_k \Bigg\} \end{aligned} \tag{38}$$

For quick reference, the last minimization term in (38), related to phase angles, is expressed in a compact way as $\phi_\theta(\boldsymbol{\Pi})$. In addition, using a component-wise formulation for the other terms making up $\phi(\boldsymbol{\Pi})$, Eq. (38) is equivalently cast as:

$$\begin{aligned}\phi(\boldsymbol{\Pi}) = & -\sum_{l\in\mathcal{L}} \left[\left(\pi_{l0}^{f+} - \pi_{l0}^{f-}\right) F_l^{ss} + \sum_{k\in\mathcal{K}} \left(\pi_{lk}^{f+} - \pi_{lk}^{f-}\right) F_l^{st}\right] \\ & - \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x}_i\in\mathcal{X}_i} \Bigg[\left(\pi_{b(i)0} - c_i\right) g_{i0} - q_i^{g,up} r_i^{g,up} - q_i^{g,dn} r_i^{g,dn} \\ & + \sum_{k\in\mathcal{K}} \psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) \Bigg] - \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y}_j\in\mathcal{Y}_j} \Bigg[\left(w_j - \pi_{b(j)0}\right) d_{j0} \\ & - q_j^{d,up} r_j^{d,up} - q_j^{d,dn} r_j^{d,dn} - \sum_{k\in\mathcal{K}} \psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) \Bigg] - \phi_\theta(\boldsymbol{\Pi}) \end{aligned} \tag{39}$$

where $\boldsymbol{x}_i = \{g_{i0}, r_i^{g,up}, r_i^{g,dn}\}$ and $\mathcal{X}_i$ is the subset of constraints related to $\boldsymbol{x}_i$, i.e., constraints (26)–(29) and $\boldsymbol{x}_i \geq \boldsymbol{0}$. Analogously, $\boldsymbol{y}_j = \{d_{j0}, r_j^{d,up}, r_j^{d,dn}\}$ and $\mathcal{Y}_j$ represents the subset of constraints related to $\boldsymbol{y}_j$, i.e., constraints (32)–(35) and $\boldsymbol{y}_j \geq \boldsymbol{0}$. Additionally, $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ respectively represent the revenue fraction earned by generator i and the payment fraction of consumer j under contingency k due to their best response against post-contingency Lagrange multiplier π_{bk}.

$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ depends on the generator availability status a_{ik}^g and on the pre-contingency dispatch and reserve schedule $\boldsymbol{x}_i$:

$$\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \max_{g_{ik}\geq 0} \ \pi_{b(i)k} g_{ik} \tag{40}$$

subject to:

$$g_{ik} \leq a_{ik}^g (g_{i0} + r_i^{g,up}) \tag{41}$$

$$g_{ik} \geq a_{ik}^g (g_{i0} - r_i^{g,dn}) \tag{42}$$

Analogously, $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ depends on the pre-contingency dispatch and reserve schedule $\boldsymbol{y}_j$:

$$\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \min_{d_{jk}\geq 0} \ \pi_{b(j)k} d_{jk} \tag{43}$$

subject to:

$$d_{jk} \geq d_{j0} - r_j^{d,up} \tag{44}$$

$$d_{jk} \leq d_{j0} + r_j^{d,dn} \tag{45}$$

The optimal solutions to problems (40)–(42) and (43)–(45) give rise to three possible outcomes for $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$:

1. If π_{bk} is positive, g_{ik}^* and d_{jk}^* will be equal to the upper and lower bounds respectively set in (41) and (44), i.e., all generators and consumers at bus b will be willing to deploy as much up-spinning reserve as possible under contingency k. As a result:
   - $g_{ik}^* = a_{ik}^g (g_{i0} + r_i^{g,up})$ and $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g (g_{i0} + r_i^{g,up})$
   - $d_{jk}^* = d_{j0} - r_j^{d,up}$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k} (d_{j0} - r_j^{d,up})$
2. If π_{bk} is negative, g_{ik}^* and d_{jk}^* will be equal to the lower and upper bounds respectively set in (42) and (45), i.e., all generators and consumers at bus b will be willing to deploy as much down-spinning reserve as possible under contingency k. As a result:
   - $g_{ik}^* = a_{ik}^g (g_{i0} - r_i^{g,dn})$ and $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i) = \pi_{b(i)k} a_{ik}^g (g_{i0} - r_i^{g,dn})$
   - $d_{jk}^* = d_{j0} + r_j^{d,dn}$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j) = \pi_{b(j)k} (d_{j0} + r_j^{d,dn})$
3. If π_{bk} equals zero, $\psi_{ik}^g(\pi_{b(i)k}, \boldsymbol{x}_i)$ and $\psi_{jk}^d(\pi_{b(j)k}, \boldsymbol{y}_j)$ will be both equal to zero for the market agents at bus b.

Moreover, defining $\pi^{+}_{bk} = \max\{\pi_{bk}, 0\}$ and $\pi^{-}_{bk} = -\min\{\pi_{bk}, 0\}$, and using the results presented above, $\psi^{g}_{ik}(\pi_{b(i)k}, \boldsymbol{x_i})$ and $\psi^{d}_{jk}(\pi_{b(j)k}, \boldsymbol{y_j})$ can be expressed as:

$$\psi^{g}_{ik}(\pi_{b(i)k}, \boldsymbol{x_i}) = \pi_{b(i)k} a^{g}_{ik} g_{i0} + \pi^{+}_{b(i)k} a^{g}_{ik} r^{g,up}_{i} + \pi^{-}_{b(i)k} a^{g}_{ik} r^{g,dn}_{i} \tag{46}$$

$$\psi^{d}_{jk}(\pi_{b(j)k}, \boldsymbol{y_j}) = \pi_{b(j)k} d_{j0} - \pi^{+}_{b(j)k} r^{d,up}_{j} - \pi^{-}_{b(j)k} r^{d,dn}_{j} \tag{47}$$

According to the strong duality theorem (Conejo et al., 1999), the minimization problem (19)–(35) and the maximization of the Lagrangian dual function (37) over $\boldsymbol{\Pi}$ yield identical values for their respective objective functions. Additionally, at the optimal dual solution $\boldsymbol{\Pi}^{*}$, the terms associated with sign-unconstrained variables $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta_k}$, represented by $\phi_{\theta}(\boldsymbol{\Pi}^{*})$, are equal to zero (Bertsimas & Tsitsiklis, 1997).

Moreover, note that $\boldsymbol{\pi}^{f+}_{0}, \boldsymbol{\pi}^{f+}_{\boldsymbol{k}} \geq \mathbf{0}$ and $\boldsymbol{\pi}^{f-}_{0}, \boldsymbol{\pi}^{f-}_{\boldsymbol{k}} \leq \mathbf{0}$. The complementary slackness condition further ensures that, at the optimal dual solution, entry-wise products $\boldsymbol{\pi}^{f+*}_{0}\boldsymbol{\pi}^{f-*}_{0}$ and $\boldsymbol{\pi}^{f+*}_{\boldsymbol{k}}\boldsymbol{\pi}^{f-*}_{\boldsymbol{k}}, \forall k \in \mathcal{K}$, are all equal to $\mathbf{0}$. Thus, defining $\boldsymbol{\pi}^{f*}_{0} = \boldsymbol{\pi}^{f+*}_{0} + \boldsymbol{\pi}^{f-*}_{0}$ and $\boldsymbol{\pi}^{f*}_{\boldsymbol{k}} = \boldsymbol{\pi}^{f+*}_{\boldsymbol{k}} + \boldsymbol{\pi}^{f-*}_{\boldsymbol{k}}$, the following equalities hold:

$$\boldsymbol{\pi}^{f+*}_{0} - \boldsymbol{\pi}^{f-*}_{0} = |\boldsymbol{\pi}^{f*}_{0}| \tag{48}$$

$$\boldsymbol{\pi}^{f+*}_{\boldsymbol{k}} - \boldsymbol{\pi}^{f-*}_{\boldsymbol{k}} = |\boldsymbol{\pi}^{f*}_{\boldsymbol{k}}|, \forall k \in \mathcal{K} \tag{49}$$

Consequently, Eq. (39) at the optimal solution is recast as Eq. (50) by using Eqs. (46)–(49), dropping $\phi_{\theta}(\boldsymbol{\Pi}^{*})$, and replacing F^{st}_{l} by $\gamma_l F^{ss}_{l}$:

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l \right) F^{ss}_{l} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi^{*}_{b(i)0} + \sum_{k\in\mathcal{K}} \pi^{*}_{b(i)k} a^{g}_{ik} - c_i \right) g_{i0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(i)k} a^{g}_{ik} - q^{g,up}_{i} \right) r^{g,up}_{i} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(i)k} a^{g}_{ik} - q^{g,dn}_{i} \right) r^{g,dn}_{i} \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi^{*}_{b(j)0} - \sum_{k\in\mathcal{K}} \pi^{*}_{b(j)k} \right) d_{j0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(j)k} - q^{d,up}_{j} \right) r^{d,up}_{j} \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(j)k} - q^{d,dn}_{j} \right) r^{d,dn}_{j} \right]
\end{aligned} \tag{50}$$

Similar to the approach described in Section 2, Eq. (50) can be recast by implicitly considering generation availability statuses a^{g}_{ik} in set $\mathcal{K}^{OFF}_{i}$ as follows:

$$\begin{aligned}
\phi\left(\boldsymbol{\Pi}^{*}\right) = &- \sum_{l\in\mathcal{L}} \left(\left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l \right) F^{ss}_{l} \\
&- \sum_{i\in\mathcal{I}} \max_{\boldsymbol{x_i}\in\mathcal{X}_i} \left[\left(\pi^{*}_{b(i)0} + \sum_{k\in\mathcal{K}} \pi^{*}_{b(i)k} - c_i \right) g_{i0} \right. \\
&+ \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(i)k} - q^{g,up}_{i} \right) r^{g,up}_{i} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(i)k} - q^{g,dn}_{i} \right) r^{g,dn}_{i} \\
&\left. - \sum_{k\in\mathcal{K}^{OFF}_{i}} \left(\pi^{*}_{b(i)k} g_{i0} + \pi^{+*}_{b(i)k} r^{g,up}_{i} + \pi^{-*}_{b(i)k} r^{g,dn}_{i} \right) \right] \\
&- \sum_{j\in\mathcal{J}} \max_{\boldsymbol{y_j}\in\mathcal{Y}_j} \left[\left(w_j - \pi^{*}_{b(j)0} - \sum_{k\in\mathcal{K}} \pi^{*}_{b(j)k} \right) d_{j0} \right. \\
&\left. + \left(\sum_{k\in\mathcal{K}} \pi^{+*}_{b(j)k} - q^{d,up}_{j} \right) r^{d,up}_{j} + \left(\sum_{k\in\mathcal{K}} \pi^{-*}_{b(j)k} - q^{d,dn}_{j} \right) r^{d,dn}_{j} \right]
\end{aligned} \tag{51}$$

The summation in the first term in the right-hand side of (51) represents the transmission congestion rent, which is made up of the revenues collected by transmission lines. Note that transmission lines are compensated through the Lagrange multipliers associated with the constraints bounding line power flows. More specifically, transmission prices result from the summation of $\left|\pi^{f*}_{l0}\right|$ and $\sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l$.

The total profit of generator i is characterized by the four terms within the first pair of square brackets in (51). The first term is the energy profit, from which the energy price at bus b is defined as $\pi^{*}_{b0} + \sum_{k\in\mathcal{K}} \pi^{*}_{bk}$, as done in the benchmark pricing system (Table 11). Interestingly, the three other generator profit terms involve salient features compared to the benchmark pricing system. First, from the second and third terms respectively related to up- and down-spinning reserves, separate prices are defined. Thus, up- and down-spinning reserve prices at bus b are equal to $\sum_{k\in\mathcal{K}} \pi^{+*}_{bk}$ and $\sum_{k\in\mathcal{K}} \pi^{-*}_{bk}$, respectively. Additionally, the fourth generator profit term models the security charge.

Finally, the terms within the second pair of square brackets in (51) represent the profit of consumer j associated with energy consumption and up- and down-spinning reserve procurement. As can be observed, the above price definitions also hold for the corresponding consumer profit terms.

It is worth noting that the maximization of the individual profit terms in (51) backs the incentive compatibility of the proposed framework.

Tables 22, 23, and 24 present the proposed pricing scheme and the corresponding generation and consumer settlements, respectively.

Tables 25–29 summarize the results from the application of the proposed pricing scheme to the two-bus example examined in Section 3.2.

As can be seen in Table 25, the proposed nodal energy prices are the same as those attained by the benchmark methodology presented

Table 22
Proposed pricing system for problem (19)–(35).

Nodal Energy Price	$p^{e}_{b} = \pi^{*}_{b0} + \sum_{k\in\mathcal{K}} \pi^{*}_{bk}$
Nodal Up-Spinning Reserve Price	$p^{up}_{b} = \sum_{k\in\mathcal{K}} \pi^{+*}_{bk}$
Nodal Down-Spinning Reserve Price	$p^{dn}_{b} = \sum_{k\in\mathcal{K}} \pi^{-*}_{bk}$
Generation Security Charge	$C^{g,s}_{i} = \sum_{k\in\mathcal{K}^{OFF}_{i}} \left(\pi^{*}_{b(i)k} g_{i0} + \pi^{+*}_{b(i)k} r^{g,up}_{i} + \pi^{-*}_{b(i)k} r^{g,dn}_{i} \right)$
Transmission Price	$p^{f}_{l} = \left|\pi^{f*}_{l0}\right| + \sum_{k\in\mathcal{K}} \left|\pi^{f*}_{lk}\right| \gamma_l$

Table 23
Generation settlement for problem (19)–(35) according to the proposed pricing system.

Generation Energy Revenue	$R^{g,e}_{i} = p^{e}_{b(i)} g_{i0}$
Generation Up-Spinning Reserve Revenue	$R^{g,up}_{i} = p^{up}_{b(i)} r^{g,up}_{i}$
Generation Down-Spinning Reserve Revenue	$R^{g,dn}_{i} = p^{dn}_{b(i)} r^{g,dn}_{i}$
Generation Total Revenue	$R^{g,t}_{i} = R^{g,e}_{i} + R^{g,up}_{i} + R^{g,dn}_{i} - C^{g,s}_{i}$
Generation Energy Cost	$C^{g,e}_{i} = c_i g_{i0}$
Generation Up-Spinning Reserve Cost	$C^{g,up}_{i} = q^{g,up}_{i} r^{g,up}_{i}$
Generation Down-Spinning Reserve Cost	$C^{g,dn}_{i} = q^{g,dn}_{i} r^{g,dn}_{i}$
Generation Total Cost	$C^{g,t}_{i} = C^{g,e}_{i} + C^{g,up}_{i} + C^{g,dn}_{i}$
Generation Profit	$Profit^{g}_{i} = R^{g,t}_{i} - C^{g,t}_{i}$

Table 24
Consumer settlement for problem (19)–(35) according to the proposed pricing system.

Consumer Payment for Energy	$CP^{e}_{j} = p^{e}_{b(j)} d_{j0}$
Consumer Up-Spinning Reserve Revenue	$R^{d,up}_{j} = p^{up}_{b(j)} r^{d,up}_{j}$
Consumer Down-Spinning Reserve Revenue	$R^{d,dn}_{j} = p^{dn}_{b(j)} r^{d,dn}_{j}$
Consumer Utility	$U^{d}_{j} = w_j d_{j0}$
Consumer Up-Spinning Reserve Cost	$C^{d,up}_{j} = q^{d,up}_{j} r^{d,up}_{j}$
Consumer Down-Spinning Reserve Cost	$C^{d,dn}_{j} = q^{d,dn}_{j} r^{d,dn}_{j}$
Consumer Total Cost	$C^{d,t}_{j} = C^{d,up}_{j} + C^{d,dn}_{j}$
Consumer Payment	$CP_j = CP^{e}_{j} - R^{d,up}_{j} - R^{d,dn}_{j}$
Consumer Profit	$Profit^{d}_{j} = U^{d}_{j} - C^{d,t}_{j} - CP_j$

Table 25
Two-bus example – Proposed nodal energy and reserve prices ($/MWh).

b	Nodal Energy Price, p_b^e	Nodal Up-Spinning Reserve Price, p_b^{up}	Nodal Down-Spinning Reserve Price, p_b^{dn}
1	200	180	0
2	100	85	5

Table 26
Two-bus example – Values of $\boldsymbol{\pi}_0^{f*}$, $\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$, and transmission price ($/MWh).

l	$\boldsymbol{\pi}_0^{f*}$	$\boldsymbol{\pi}_{\boldsymbol{k}}^{f*}$				Transmission Price, p_l^f
		Outage of Generator 1	Outage of Generator 2	Outage of Generator 3	Outage of Line 1–2	
1–2	0	95	0	0	–	95

Table 27
Two-bus example – Proposed generation settlement ($).

i	$R_i^{g,e}$	$R_i^{g,up}$	$R_i^{g,dn}$	$C_i^{g,s}$	$R_i^{g,t}$	$C_i^{g,e}$	$C_i^{g,up}$	$C_i^{g,dn}$	$C_i^{g,t}$	$Profit_i^g$
1	15,000	0	0	13,500	1,500	1,500	0	0	1,500	0
2	3,000	2,550	25	0	5,575	1,500	150	25	1,675	3,900
3	1,500	2,975	0	0	4,475	1,500	350	0	1,850	2,625
Total	**19,500**	**5,525**	**25**	**13,500**	**11,550**	**4,500**	**500**	**25**	**5,025**	**6,525**

Table 28
Two-bus example – Proposed consumer settlement ($).

j	CP_j^e	$R_j^{d,up}$	$R_j^{d,dn}$	CP_j	U_j^d	$C_j^{d,up}$	$C_j^{d,dn}$	$C_j^{d,t}$	$Profit_j^d$
1	16,000	1,800	0	14,200	16,000	1,500	0	1,500	300
2	4,000	0	0	4,000	6,000	0	0	0	2,000
Total	**20,000**	**1,800**	**0**	**18,200**	**22,000**	**1,500**	**0**	**1,500**	**2,300**

Table 29
Two-bus example – Proposed system settlement ($).

Generation Revenue	11,550
Transmission Revenue	6,650
Consumer Payment	18,200
Balance	0

in Arroyo and Galiana (2005), thereby leading to identical energy revenues for generators and energy payments by consumers. By contrast, the use of different nodal prices for up- and down-spinning reserves (Table 25) constitutes a significant departure from the single nodal price defined in the benchmark pricing system for both services, thus yielding substantial differences for the corresponding revenues. For this particular example, the comparison of Tables 27 and 19 shows that generators 2 and 3 increase their up-spinning reserve revenues by 6.25% whereas the down-spinning reserve revenue of generator 2 decreases by a factor of 16. Additionally, as generator 1 is responsible for the need for up-spinning reserve, a security charge is levied on this generator.

As for consumers, the settlement remains unaltered for this particular example (Tables 28 and 20). Note that the only load contributing to security is load 1 at bus 1, in the form of up-spinning reserve. For this bus, the proposed up-spinning reserve price happens to be the same as the security price resulting from the benchmark method, thereby giving rise to identical revenues.

Moreover, unlike the benchmark pricing system, line congestion under contingency is acknowledged by the proposed transmission price p_l^f (Table 26) and the corresponding transmission revenue, $p_l^f F_l^{ss}$, which, for this case, amounts to $95 \times 70 = \$6,650$.

As expected, the total generation profit, the total consumer profit, and the transmission revenue sum up the aforementioned optimal social welfare, i.e., $15,475. Remarkably, different from the benchmark pricing scheme, the proposed method renders the total consumer payment equal to the sum of the generation revenue and the transmission revenue (Table 29), as is desirable. These results provide empirical support for the cost recovery and system revenue adequacy featured by the proposed pricing scheme for the market-clearing problem analyzed in this section. The corresponding extensions of Theorems 1 and 2 and their proofs are provided next.

Theorem 3 (*Cost Recovery*). *The proposed pricing framework ensures that generators' and consumers' profits are non-negative.*

Proof. Let us rewrite the LD function in Eq. (51) as follows:

$$\begin{aligned}\phi\left(\boldsymbol{\Pi}^*\right) = &-\sum_{l\in\mathcal{L}}\left(\left|\pi_{l0}^{f*}\right| + \sum_{k\in\mathcal{K}}\left|\pi_{lk}^{f*}\right|\gamma_l\right)F_l^{ss} \\ &- \sum_{i\in\mathcal{I}}\max_{\boldsymbol{x_i}\in\mathcal{X}_i}\left[Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)\right] \\ &- \sum_{j\in\mathcal{J}}\max_{\boldsymbol{y_j}\in\mathcal{Y}_j}\left[Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)\right]\end{aligned} \tag{52}$$

The terms $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)$ respectively represent the profits of generator i and consumer j. It should be noted that $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j}\right)$ are linear objective functions respectively maximized over $\boldsymbol{x_i}$ and $\boldsymbol{y_j}$ with $\boldsymbol{x_i} \geq \boldsymbol{0}$ and $\boldsymbol{y_j} \geq \boldsymbol{0}$. Therefore, at the optimal solution, $Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i^*}\right)$ and $Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j^*}\right)$ are both non-negative. In other words, if $\pi_{b(i)0}^*$ and $\pi_{b(i)k}^*$ fail to cover generator i's costs, the production and reserve contributions of this generator will be 0, i.e., $g_{i0}^* = r_i^{g,up*} = r_i^{g,dn*} = 0$. Analogously, if $\pi_{b(j)0}^*$ and $\pi_{b(j)k}^*$ are not profitable for consumer j, the consumption and reserve contributions of this consumer will be 0, i.e., $d_{j0}^* = r_j^{d,up*} = r_j^{d,dn*} = 0$. Therefore:

$$Profit_i^g\left(\boldsymbol{\Pi}_{\boldsymbol{b(i)}}^*, \boldsymbol{x_i^*}\right) \geq 0, \ \forall i \in \mathcal{I} \tag{53}$$

$$Profit_j^d\left(\boldsymbol{\Pi}_{\boldsymbol{b(j)}}^*, \boldsymbol{y_j^*}\right) \geq 0, \ \forall j \in \mathcal{J} \tag{54}$$

Thus, the theorem holds. □

Theorem 4 (*System Revenue Adequacy*). *The total payment by consumers is equal to the sum of generation and transmission revenues.*

Proof. By the strong duality theorem, the optimal value of the primal objective function in (19) is equal to the optimal value of the LD function in (51). Thus, at the optimal solution $\boldsymbol{x}^*$, $\boldsymbol{y}^*$, and $\boldsymbol{\Pi}^*$, we have:

$$\begin{aligned}&\sum_{i\in\mathcal{I}} c_i g_{i0}^* + \sum_{i\in\mathcal{I}} q_i^{g,up} r_i^{g,up*} + \sum_{i\in\mathcal{I}} q_i^{g,dn} r_i^{g,dn*} - \sum_{j\in\mathcal{J}} w_j d_{j0}^* \\ &+ \sum_{j\in\mathcal{J}} q_j^{d,up} r_j^{d,up*} + \sum_{j\in\mathcal{J}} q_j^{d,dn} r_j^{d,dn*} = \\ &-\sum_{l\in\mathcal{L}}\left(|\pi_{l0}^{f*}| + \sum_{k\in\mathcal{K}}|\pi_{lk}^{f*}|\gamma_l\right)F_l^{ss} - \sum_{i\in\mathcal{I}}\Bigg[\left(\pi_{b(i)0}^* + \sum_{k\in\mathcal{K}}\pi_{b(i)k}^* - c_i\right)g_{i0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{+*} - q_i^{g,up}\right)r_i^{g,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(i)k}^{-*} - q_i^{g,dn}\right)r_i^{g,dn*} \\ &- \sum_{k\in\mathcal{K}_i^{\mathrm{OFF}}}\left(\pi_{b(i)k}^* g_{i0}^* + \pi_{b(i)k}^{+*} r_i^{g,up*} + \pi_{b(i)k}^{-*} r_i^{g,dn*}\right)\Bigg] \\ &- \sum_{j\in\mathcal{J}}\Bigg[\left(w_j - \pi_{b(j)0}^* - \sum_{k\in\mathcal{K}}\pi_{b(j)k}^*\right)d_{j0}^* \\ &+ \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{+*} - q_j^{d,up}\right)r_j^{d,up*} + \left(\sum_{k\in\mathcal{K}}\pi_{b(j)k}^{-*} - q_j^{d,dn}\right)r_j^{d,dn*}\Bigg]\end{aligned} \tag{55}$$

By canceling out identical terms in both sides of Eq. (55) and rearranging terms, we simplify the expression as follows:

$$\begin{aligned}
&\sum_{j\in\mathcal{J}}\left[\left(\pi^{*}_{b(j)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(j)k}\right)d^{*}_{j0}-\sum_{k\in\mathcal{K}}\pi^{+*}_{b(j)k}r^{d,up*}_{j}-\sum_{k\in\mathcal{K}}\pi^{-*}_{b(j)k}r^{d,dn*}_{j}\right]=\\
&\sum_{l\in\mathcal{L}}\left(|\pi^{f*}_{l0}|+\sum_{k\in\mathcal{K}}|\pi^{f*}_{lk}|\gamma_{l}\right)F^{ss}_{l}+\sum_{i\in\mathcal{I}}\left[\left(\pi^{*}_{b(i)0}+\sum_{k\in\mathcal{K}}\pi^{*}_{b(i)k}\right)g^{*}_{i0}\right.\\
&\quad+\sum_{k\in\mathcal{K}}\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\sum_{k\in\mathcal{K}}\pi^{-*}_{b(i)k}r^{g,dn*}_{i}\\
&\quad\left.-\sum_{k\in\mathcal{K}^{\mathrm{OFF}}_{i}}\left(\pi^{*}_{b(i)k}g^{*}_{i0}+\pi^{+*}_{b(i)k}r^{g,up*}_{i}+\pi^{-*}_{b(i)k}r^{g,dn*}_{i}\right)\right]
\end{aligned}\tag{56}$$

Therefore, at the optimal solution, the sum of all consumer payments equals the sum of all generator and transmission line revenues. □

4. 300-Bus case study

To assess the feasibility and robustness of the proposed pricing framework, we examine a larger case study based on a modified version of the standard IEEE 300-bus test system (Matpower, 2026). This system comprises 300 buses, 69 generators, and 822 transmission lines. Thus, 891 contingency states are considered, corresponding to the outage of every single line and every single generator. For the sake of reproducibility, system data are available in Ribeiro et al. (2025).

The optimal market-clearing solution attained by the simplex algorithm of CPLEX features a total value of the objective function equal to −$8,087,856.74. In the pre-contingency state, the market-clearing process balances total generation and price-responsive demand at 17,991.93 MW. This value represents the total power cleared in the system, reflecting both the aggregate supply and the corresponding consumption level.

For this case, both the proposed approach and the benchmark pricing system described in Arroyo and Galiana (2005) yield identical energy prices, which are shown in Fig. 2. To provide insight into the effect of contingencies on pricing, we have also implemented the contingency-unconstrained market-clearing counterpart associated with an $N-0$ security criterion, for which no reserves are needed. The optimal $N-0$ dispatch, which is 85% cheaper, as expected, yields no transmission congestion and, hence, no congestion rents, giving rise to a uniform nodal energy price of $500/MWh. The comparison of this result with the nodal energy prices shown in Fig. 2 reveals that neglecting security does not necessarily lead to lower energy prices, as also reported in Arroyo and Galiana (2005).

In contrast to energy, reserves are priced differently by both pricing frameworks. It should be noted that the up- and down-spinning reserve prices identified by the proposed scheme all are non-negative, whereas the benchmark method used for assessment purposes gives rise to negative security prices for some buses. Table 30 summarizes the results for the units with the largest discrepancies in down-spinning reserve remuneration. To facilitate a direct comparison, results from both methods are reported side by side, with their respective framework labels indicated in the first row of the table. More specifically, for each unit, this table lists (1) the allocated down-spinning reserve, (2) the corresponding nodal security price and the down-spinning reserve revenue both resulting from the benchmark scheme, and (3) the respective nodal down-spinning reserve price and down-spinning reserve revenue determined by the proposed approach. In the benchmark pricing method, negative security prices arise from their definition as the sum of Lagrange multipliers across all contingency states (Table 11). When the system is primarily constrained by the need to absorb excess generation, i.e., when downward reserve requirements dominate, these multipliers become negative, potentially leading to negative security prices. Table 30 shows that, under the benchmark scheme, units 11 and 31 receive significantly higher remuneration, while units 16, 28, and 56 collect negative revenues. These results underscore the critical importance of one of the salient aspects of the proposed approach, namely separately pricing upward and downward reserves according to their respective opportunity costs, thereby avoiding the distortions associated with a single aggregated security price.

Table 30
300-bus example – Generation down-spinning reserve results.

i	$r^{g,dn}_{i}$ (MW)	Benchmark Pricing System		Proposed Pricing System	
		$p^{s}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)	$p^{dn}_{b(i)}$ ($/MWh)	$R^{g,dn}_{i}$ ($)
11	255.45	34.12	8,715.75	15.00	3,831.73
31	200.00	33.00	6,600.00	17.00	3,400.00
16	318.32	−6.00	−1,909.92	6.00	1,909.92
56	159.77	−7.69	−1,002.20	12.00	1,564.33
28	72.50	−6.00	−958.60	6.00	958.60

Fig. 3 shows the impact of both pricing systems on generation profits. Note that, for each generator, the profit difference depicted in Fig. 3 is computed as the relative difference between the profit attained under the proposed pricing scheme and the profit determined by the benchmark pricing system, expressed as a percentage of the benchmark profit. For expository purposes, generators are sorted along the horizontal axis of Fig. 3 in descending order of this percentage difference. Moreover, generators with negligible benchmark profits are excluded from this figure, as small absolute differences would otherwise translate into disproportionately large percentage variations. According to Fig. 3, substantial profit disparities arise, including percentage differences close to −100%, which correspond to generators with non-negative profits under the proposed pricing scheme and negative profits under the benchmark framework. Thus, this example is useful to illustrate another key advantage of the proposed pricing method, namely cost recovery. As expected, the proposed approach yields non-negative profits for all generators, unlike the benchmark pricing system, which leads to negative profits for 24 units totaling −$24,529.63.

Fig. 4 depicts the security charge along with the energy and reserve revenues identified by the proposed pricing mechanism for each generator. In order to better illustrate the effect of the cost-causation principle, pre-contingency generation dispatch levels from highest to lowest are provided in the horizontal axis. It should be noted that generators operating at higher pre-contingency dispatch levels tend to face higher security charges, since their outage imposes greater stress on the system, thereby requiring larger reserve procurement. Relevant exceptions to this trend are shown in Fig. 4, though, mainly stemming from the crucial impact of the network topology on the extent to which a generator's failure restricts system operation.

As for transmission, the proposed approach identified 106 lines with non-zero transmission revenues. It is worth highlighting that 90 of those lines, i.e., 84.9%, featured flow capacity constraints that are non-binding in the pre-contingency state but become binding under contingency. This result further illustrates that the contingency-constrained market-clearing model under consideration ensures that the transmission system has capacity reserved to ensure feasibility under contingency conditions. Fig. 5 presents the non-zero transmission revenues identified by the proposed framework for such pre-contingency uncongested lines. For illustration purposes, the horizontal axis displays the corresponding line indices, ordered by revenue magnitude. Notably, these lines collect revenues by operating in the pre-contingency state below their capacity, which is then fully utilized to accommodate feasible post-contingency flows, thereby ensuring reserve deliverability throughout the network under contingency conditions. Thus, in contrast to the benchmark pricing system, the proposed pricing scheme captures the system's opportunity costs associated with the scarce capacity of these lines and, consequently, extends the concept of transmission rent allocation to contingency states under the general marginal cost principle.

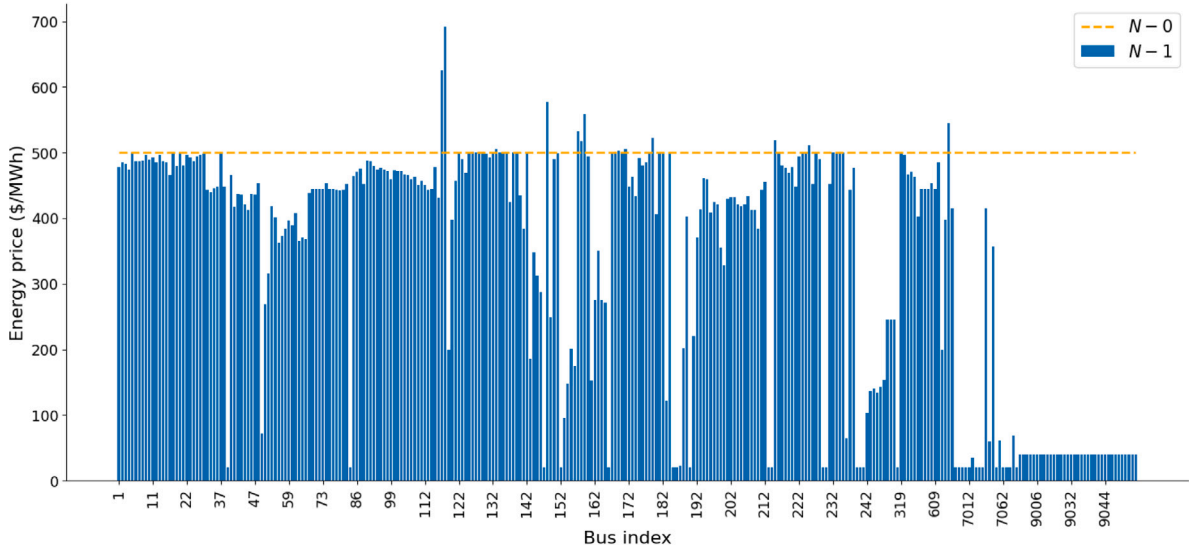


Fig. 2. 300-bus example – Nodal energy prices.

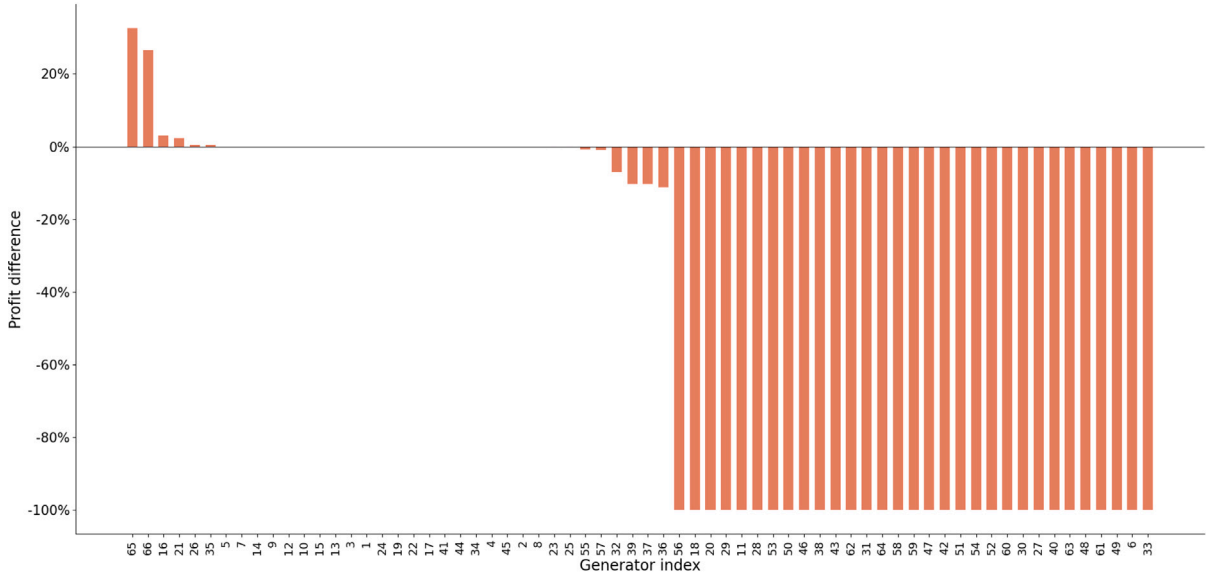


Fig. 3. 300-bus example – Percentage differences in generation profits.

Finally, Table 31 provides the system settlement for both pricing schemes. According to Table 31, consumers pay $9,957.69 less under the proposed pricing scheme than under the benchmark framework, while the total generation revenue is $208,229.00 lower, yet no unit incurs negative profit. These results are particularly relevant from the perspective of consumers. As previously discussed, the benchmark pricing scheme yields negative profits for a subset of generators, which would need to be compensated through payments in addition to those resulting from marginal pricing (e.g., make-whole payments), further increasing consumers' total payments beyond the market settlement given in Table 31. On the other hand, under the proposed pricing scheme, consumers pay less without requiring any payments beyond those associated with the marginal prices. Furthermore, the proposed pricing scheme also benefits the transmission system. As reported in Table 31, in the absence of transmission prices and the associated revenues, the benchmark approach produces a surplus that falls short by $198,271.31 of the total transmission revenue collected under the proposed scheme. In addition, as also shown in Table 31, the improvements introduced in the proposed pricing scheme allow the attainment of system revenue adequacy, unlike the benchmark method. Note that, as a desirable result, the proposed settlement yields a net balance of $0.00, i.e., the total consumer payment is fully distributed to generators and transmission assets, with neither surplus nor deficit.

5. Conclusion

This paper has presented a new causation-based pricing framework as an alternative to a previously reported scheme. Major salient features include explicitly defining prices for up-spinning reserves, down-spinning reserves, and transmission services, along with a novel security charge mechanism. These additions ensure a more comprehensive allocation of the costs associated with operating reserves and system security, resulting in efficient market operations. The proposed approach is rigorously grounded in the Lagrangian dual function, leveraging Lagrange multipliers to establish nodal prices and security charges. More importantly, the proposed pricing scheme yields a market settlement featuring two relevant and desirable properties, namely cost recovery

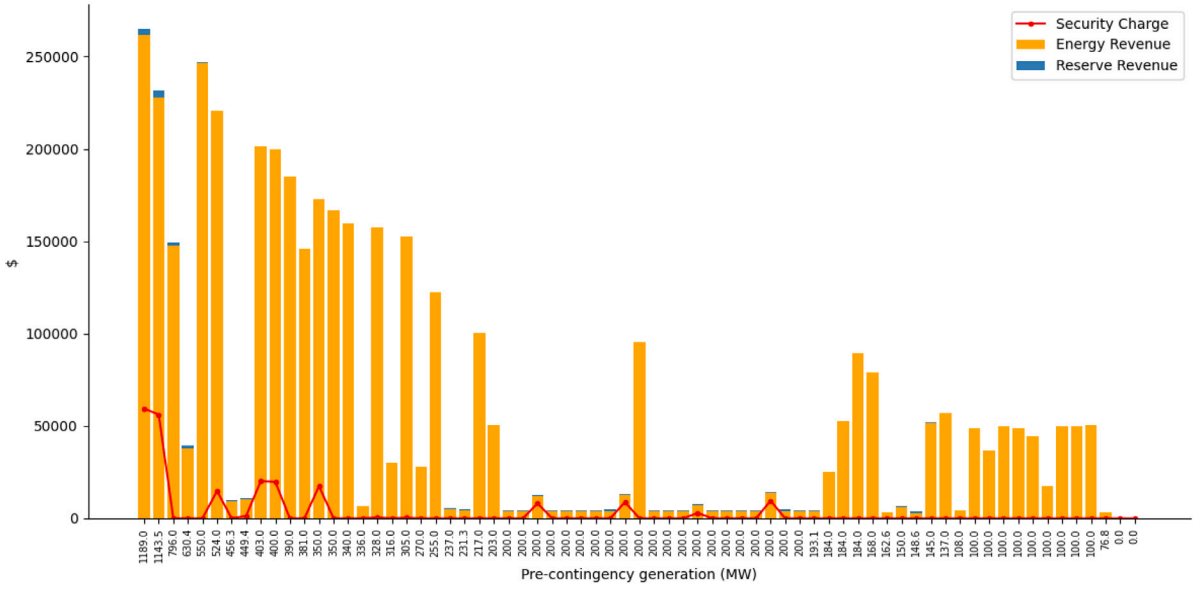


Fig. 4. 300-bus example – Generation revenues and security charges for the proposed framework.

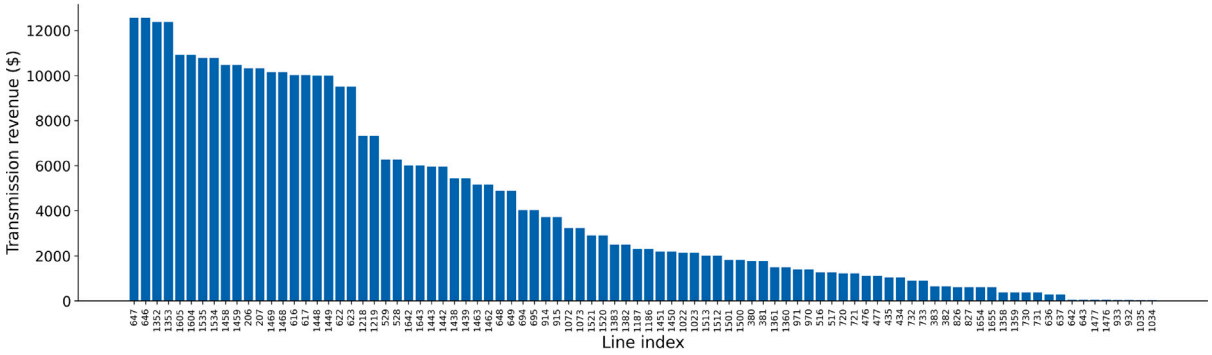


Fig. 5. 300-bus example – Non-zero transmission revenues identified by the proposed framework for the pre-contingency uncongested lines.

Table 31
300-bus example – System settlement.

Framework	Consumer Payment ($)	Generation Revenue ($)	Transmission Revenue ($)	Balance ($)
Benchmark Pricing System	7,464,201.68	4,048,997.62	0.00	3,415,204.06
Proposed Pricing System	7,454,243.99	3,840,768.62	3,613,475.37	0.00
Difference	9,957.69	208,229.00	−3,613,475.37	3,415,204.06

and system revenue adequacy, avoiding the need for any additional payments beyond those resulting from marginal pricing.

Numerical results corroborate the findings in the related literature, demonstrating that generators scheduled with higher pre-contingency output tend to bear higher security charges due to their significant contribution to the system-wide reserve requirements under contingency. Interestingly, based on the cost-causation principle, these charges remain consistent with the uniform pricing system largely used for energy and reserves. Additionally, the newly proposed pricing framework involves transmission assets, remunerating them based on energy and reserve utilization across all contingency states. This unification guarantees a consistent and transparent cost allocation framework. Overall, the proposed pricing system provides a sound extension of transmission rent allocation by capturing system's opportunity costs arising from scarce transmission capacity and consistently accounting for contingency states under the marginal cost principle.

The main limitation of the proposed approach is the adoption of a deterministic, single-period model relying on a convex characterization of system operation. Thus, this work paves the way for future research in this field. One potential direction is the extension to explicitly consider inter-temporal aspects and nonconvexities in market operations such as unit commitment constraints, which require binary variables, and an ac power flow model. The challenges associated with primal degeneracy, i.e., the existence of multiple vectors of Lagrange multipliers for the same optimal market-clearing solution, represent another interesting avenue of research. Moreover, the framework could be enhanced to model more complex, multi-stage security criteria, such as the $N-1-1$ standard. This modeling feature would involve incorporating different reserve products with varying ramp rates and activation times to represent the distinct requirements for immediate contingency response and subsequent system restoration. Furthermore, having established a consistent deterministic framework, a natural next step is to incorporate the uncertainty associated with component outages and nodal injections. This extension will be particularly relevant within the

future low-inertia renewables-dominated paradigm, complementing the existing security-based approach with a more granular assessment of operational risks. In this context, the modeling framework could be generalized to represent not only full generator outages but also partial derating of generating units as well as consumer contingencies giving rise to demand security charges. Additionally, further investigation into economic incentives could provide deeper insights into how market participants, including smaller generators and consumers, strategically respond to security charges and reserve pricing while also considering the potential exercise of market power.

Finally, it is worth emphasizing that explicitly charging generators according to a security criterion may be regarded as a nontrivial departure from current market designs, as most system operators do not directly internalize such charges at the generation level. In this regard, a detailed discussion of the associated regulatory and market design implications is beyond the scope of this paper. Interestingly, the proposed framework is agnostic with respect to the ultimate allocation of these costs, which may be charged to generators or alternatively passed through to consumers according to prevailing regulatory arrangements. Nevertheless, the proposed pricing framework should be interpreted as a methodological advancement in identifying transmission and reserve price signals and components, consistently with the dual prices associated with each contingency embedded in the security criterion. In this sense, the proposed pricing system can be of great value to system operators and regulators in identifying the critical components that contribute to reserve costs and to the value of transmission for system security. Given the regulatory nature of this subject, it can be viewed as a relevant, policy-based extension of this work—including the study of alternative allocation rules, such as those based on realized profits—and is therefore left for future research.

CRediT authorship contribution statement

Luíza Ribeiro: Writing – review & editing, Writing – original draft. **Alexandre Street:** Writing – review & editing, Project administration, Methodology, Funding acquisition. **José M. Arroyo:** Writing – review & editing, Supervision. **Rodrigo Moreno:** Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Nomenclature

This section lists the main notation used throughout the paper. Note that superscript "*" stands for optimal value.

Sets and indices:

b	Bus index.
$\mathcal{I}$	Set of indices i of generators.
$\mathcal{J}$	Set of indices j of loads.
$\mathcal{K}$	Set of indices k of contingencies.
$\mathcal{K}_i^{OFF}$	Set of indices k of the contingencies involving the loss of generator i.
$\mathcal{L}$	Set of indices l of lines.
$\mathcal{X}$	Generator feasibility set.
$\mathcal{X}_i$	Subset of $\mathcal{X}$ related to generator i.
$\mathcal{Y}$	Consumer feasibility set.
$\mathcal{Y}_j$	Subset of $\mathcal{Y}$ related to consumer j.

Constants:

$\boldsymbol{\gamma}$	Vector of contingency flow-relief factors, with each element denoted by γ_l.
$\boldsymbol{A}$	Incidence matrix in the pre-contingency state.
$\boldsymbol{A_k}$	Incidence matrix under contingency k.
$\boldsymbol{a_k^g}$	Vector of generator availability statuses under contingency k, with each element denoted by a_{ik}^g.
$\boldsymbol{c}$	Vector of cost rates offered by generators to provide energy in the pre-contingency state, with each element denoted by c_i.
d	Inelastic demand.
$\overline{\boldsymbol{D}}$	Vector of upper bounds on the power consumed, with each element denoted by $\overline{D}_j$.
$\boldsymbol{F^{ss}}$	Vector of steady-state line power flow capacities, with each element denoted by F_l^{ss}.
$\boldsymbol{F^{st}}$	Vector of short-term line power flow capacities, with each element denoted by F_l^{st}.
$\overline{\boldsymbol{G}}$	Vector of generator capacities, with each element denoted by $\overline{G}_i$.
$\boldsymbol{H}$	Matrix relating power flows to nodal phase angles in the pre-contingency state.
$\boldsymbol{H_k}$	Matrix relating power flows to nodal phase angles under contingency k.
$\boldsymbol{M^d}$	Demand-bus mapping matrix.
$\boldsymbol{M^g}$	Generator-bus mapping matrix.
$\boldsymbol{q^{d,dn}}$	Vector of cost rates offered by consumers to provide down-spinning reserve, with each element denoted by $q_j^{d,dn}$.
$\boldsymbol{q^{d,up}}$	Vector of cost rates offered by consumers to provide up-spinning reserve, with each element denoted by $q_j^{d,up}$.
$\boldsymbol{q^{g,dn}}$	Vector of cost rates offered by generators to provide down-spinning reserve, with each element denoted by $q_i^{g,dn}$.
$\boldsymbol{q^{g,up}}$	Vector of cost rates offered by generators to provide up-spinning reserve, with each element denoted by $q_i^{g,up}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{d,dn}}$	Vector of upper bounds for consumer down-spinning reserves, with each element denoted by $\overline{R}_j^{d,dn}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{d,up}}$	Vector of upper bounds for consumer up-spinning reserves, with each element denoted by $\overline{R}_j^{d,up}$.
$\overline{\boldsymbol{R}}^{\boldsymbol{g,dn}}$	Vector of upper bounds for generator down-spinning reserves, with each element denoted by $\overline{R}_i^{g,dn}$.

$\overline{\boldsymbol{R}}^{g,up}$ Vector of upper bounds for generator up-spinning reserves, with each element denoted by $\overline{R}_i^{g,up}$.
$\boldsymbol{w}$ Vector of rates bid by consumers to buy energy, with each element denoted by w_j.
$\boldsymbol{X}$ Diagonal matrix of line reactances.

Variables:

$\boldsymbol{\theta_0}$ Vector of pre-contingency nodal phase angles.
$\boldsymbol{\theta_k}$ Vector of nodal phase angles under contingency k.
$\boldsymbol{d}_0$ Vector of pre-contingency nodal consumption levels, with each element denoted by d_{b0}.
$\boldsymbol{d_k}$ Vector of nodal consumption levels under contingency k, with each element denoted by d_{bk}.
$\boldsymbol{g}_0$ Vector of generator power outputs in the pre-contingency state, with each element denoted by g_{i0}.
$\boldsymbol{g_k}$ Vector of generator power outputs under contingency k, with each element denoted by g_{ik}.
$\boldsymbol{r^{d,dn}}$ Vector of down-spinning reserves provided by consumers, with each element denoted by $r_j^{d,dn}$.
$\boldsymbol{r^{d,up}}$ Vector of up-spinning reserves provided by consumers, with each element denoted by $r_j^{d,up}$.
$\boldsymbol{r^{g,dn}}$ Vector of down-spinning reserves provided by generators, with each element denoted by $r_i^{g,dn}$.
$\boldsymbol{r^{g,up}}$ Vector of up-spinning reserves provided by generators, with each element denoted by $r_i^{g,up}$.
$\boldsymbol{x}$ Vector of optimization variables related to generators.
$\boldsymbol{x_i}$ Vector of optimization variables related to generator i.
$\boldsymbol{y}$ Vector of optimization variables related to consumers.
$\boldsymbol{y_j}$ Vector of optimization variables related to consumer j.

Lagrange multipliers:

$\boldsymbol{\Pi}$ Vector of all Lagrange multipliers.
$\boldsymbol{\Pi_b}$ Subvector of $\boldsymbol{\Pi}$ related to bus b.
$\boldsymbol{\pi_0}$ Vector of Lagrange multipliers associated with the pre-contingency nodal power balance constraints, with each element denoted by π_{b0}.
π_0 Lagrange multiplier associated with the pre-contingency power balance constraint in the single-bus model.
$\boldsymbol{\pi_k}$ Vector of Lagrange multipliers associated with the nodal power balance constraints under contingency k, with each element denoted by π_{bk}.
π_k Lagrange multiplier associated with the nodal power balance constraint under contingency k in the single-bus model.
$\boldsymbol{\pi_0^{f+}}$ Vector of Lagrange multipliers associated with the lower bounds for the pre-contingency line flows, with each element denoted by π_{l0}^{f+}.
$\boldsymbol{\pi_0^{f-}}$ Vector of Lagrange multipliers associated with the upper bounds for the pre-contingency line flows, with each element denoted by π_{l0}^{f-}.
$\boldsymbol{\pi_k^{f+}}$ Vector of Lagrange multipliers associated with the lower bounds for the line flows under contingency k, with each element denoted by π_{lk}^{f+}.
$\boldsymbol{\pi_k^{f-}}$ Vector of Lagrange multipliers associated with the upper bounds for the line flows under contingency k, with each element denoted by π_{lk}^{f-}.

Functions:

$\phi(\cdot)$ Lagrangian dual function.
$\phi_\theta(\cdot)$ Term of the Lagrangian dual function associated with $\boldsymbol{\theta_0}$ and $\boldsymbol{\theta_k}$.
$\psi_{jk}^d(\cdot)$ Consumer j's payment fraction under contingency k.
$\psi_{ik}^g(\cdot)$ Generator i's revenue fraction under contingency k.

Others:

π_{bk}^+ Contribution to p_b^{up} due to contingency k.
π_{bk}^- Contribution to p_b^{dn} due to contingency k.
$\boldsymbol{\pi_0^f}$ Vector resulting from summing $\boldsymbol{\pi_0^{f+}}$ and $\boldsymbol{\pi_0^{f-}}$, with each element denoted by π_{l0}^f.
$\boldsymbol{\pi_k^f}$ Vector resulting from summing $\boldsymbol{\pi_k^{f+}}$ and $\boldsymbol{\pi_k^{f-}}$, with each element denoted by π_{lk}^f.
$b(i)$ Bus of generator i.
$b(j)$ Bus of consumer j.
$C_j^{d,dn}$ Down-spinning reserve offer cost of consumer j.
$C_j^{d,t}$ Total cost of consumer j.
$C_j^{d,up}$ Up-spinning reserve offer cost of consumer j.
$C_i^{g,dn}$ Down-spinning reserve offer cost of generator i.
$C_i^{g,e}$ Energy cost of generator i.
$C_i^{g,s}$ Security charge of generator i.
$C_i^{g,t}$ Total cost of generator i.
$C_i^{g,up}$ Up-spinning reserve offer cost of generator i.
CP Consumer payment.
CP_j Payment of consumer j.
CP_j^e Energy payment of consumer j.
p_b^{dn} Down-spinning reserve price at bus b.
p^e Energy price.
p_b^e Energy price at bus b.
p_l^f Transmission price for line l.
p^s Security price.
p_b^s Security price at bus b.
p^{up} Up-spinning reserve price.
p_b^{up} Up-spinning reserve price at bus b.
$Profit_j^d$ Profit of consumer j.
$Profit_i^g$ Profit of generator i.
$R_j^{d,dn}$ Down-spinning reserve revenue of consumer j.
$R_j^{d,up}$ Up-spinning reserve revenue of consumer j.
$R_i^{g,dn}$ Down-spinning reserve revenue of generator i.
$R_i^{g,e}$ Energy revenue of generator i.
$R_i^{g,t}$ Total revenue of generator i.
$R_i^{g,up}$ Up-spinning reserve revenue of generator i.
U_j^d Utility of consumer j.

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